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Supplementary Materials for

Imbalance in gut microbial interactions as a marker of health and disease

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The PDF file includes:

Materials and Methods
Figs. S1 to S24
Table S1
References

Other Supplementary Material for this manuscript includes the following:

MDAR Reproducibility Checklist

Materials and Methods

Measures

- α**-Diversity:** measured using the Shannon Index, defined as

$$\text{Shannon Index} = -\sum_i b_i \ln b_i \tag{S1}$$

where b_i represents the relative abundance (biomass) of type i in the model, or the relative abundance (proportion of counts) of species i in the metagenomic data.

- **Richness:** measured using the number of distinct bacterial types in the model, or taxonomic entities in the real-world data.

- **Dominance:** measured as the relative abundance of the most abundant bacterial type (model) or taxonomic entity (real-world data).

- **Simpson Index:** defined as

$$\text{Simpson} = 1 - \sum_i b_i^2 \tag{S2}$$

where b_i represents the relative abundance of bacterial type i (model) or species i (metagenomic data).

- **Pielou's evenness:** calculated as the Shannon index divided by the logarithm of richness:

$$\text{Pielou} = \frac{\text{Shannon Index}}{\ln(\text{Richness})} \tag{S3}$$

- **β-diversity:** measured using the Bray-Curtis (BC) dissimilarity index between two samples, a and d, defined as:

$$BC_{a,d} = 1 - \frac{2\sum_i \min\left(b_i^a, b_i^d\right)}{\sum_i \left(b_i^a + b_i^d\right)} \tag{S4}$$

where b_i^a and b_i^d represent, for samples a and d, the relative abundance of type i (model) or species i (metagenomic data). A value of $BC_{a,d} = 0$ indicates identical community compositions, while $BC_{a,d} = 1$ indicates completely distinct communities.

- **Number of Enzymes:** In real data, we calculated the number of enzymes as the sum of enzymes with non-zero abundance in the sample. In the model, because enzymes are present through the enzymatic cost of a pathway, we calculated a proxy for the number of enzymes, the total enzymatic cost in the sample:

$$\text{Total Enzymatic Cost} = \sum_{i\alpha\beta} P_{\alpha\beta}^{i}\, C_{\alpha\beta} \tag{S5}$$

where $P_{\alpha\beta}^{i}$ represents the pathway matrix of bacterial type i and $C_{\alpha\beta}$ the associated cost stemming from enzyme production.

- **Number of Metabolites:** Calculated as the sum of non-zero metabolites

$$\text{Number of Metabolites} = \sum_{\alpha} \theta\left(S_\alpha\right) \tag{S6}$$

where $\theta(x)$ is the Heaviside function (i.e. $\Theta(x) = 1$ if $x > 0$, and $\Theta(x) = 0$ otherwise) and S_α is the density of metabolite α, both in the model and metabolomic data.

- **Pathway fraction:** In real data, the pathway fraction is computed as the relative abundance of each metabolic pathway. In the model, we define it as:

$$\text{Pathway Fraction}_{\alpha\beta} = \frac{\sum_i P^i_{\alpha\beta}\, C_{\alpha\beta} B_i}{\sum_{i,\alpha,\beta} P^i_{\alpha\beta}\, C_{\alpha\beta} B_i} \tag{S7}$$

  This formulation incorporates both the enzymatic cost and the biomass of the species expressing each pathway, providing a proxy for the functional signal that would be obtained from pathway abundance profiles in metagenomic data.

- **Cross-feeding:** calculated as

$$c^+_{ij} = \sum_{\alpha\beta\delta} P^i_{\alpha\beta}\, P^j_{\beta\delta} r_i(S_\alpha) r_j(S_\beta) B_i B_j \tag{S8}$$

  which captures all the cross-feeding interactions from type i to type j, mediated by all possible metabolic pathways. Specifically, c^+_{ij} considers all instances where species i consumes substance S_α to generate substance S_β, which is then consumed by species j to produce any byproduct S_δ. Each interaction is weighted by the corresponding species abundances (B_i and B_j) and consumption factor r_i, the latter given by the Monod function $r(S) = r^{\max} \frac{S}{K+S}$. The matrix composed of all c^+_{ij} is always positive and non-symmetric.

- **Competition:** calculated as

$$c^-_{ij} = \sum_{\alpha\beta\delta} P^i_{\alpha\beta}\, P^j_{\alpha\delta} r_i(S_\alpha) r_j(S_\alpha) B_i B_j \tag{S9}$$

which captures the competition between types i and j arising from their shared preference for the same substance S_α. Each competitive interaction is weighted by the corresponding species abundances and consumption functions. The matrix composed using all c_{ij}^- is always positive and symmetric.

- **Net interaction:** defined as

$$\rho = \frac{\sum_{i \neq j} \left(c_{ij}^+ - c_{ij}^-\right)}{\sum_{i \neq j} \left(c_{ij}^+ + c_{ij}^-\right)} \tag{S10}$$

where the terms c_{ij}^+ and c_{ij}^- represent, respectively, the elements of the cross-feeding and competition matrices above.

- **Inferred net interaction:** defined as

$$\rho_{\text{inferred}} = \frac{\sum_{i \neq j} \left(\omega_{ij}^+ - \omega_{ij}^-\right)}{\sum_{i \neq j} \left(\omega_{ij}^+ + \omega_{ij}^-\right)} \tag{S11}$$

where ω_{ij} represent the positive (ω_{ij}^+) or negative (ω_{ij}^-) weights between types (in the model) or species (metagenomic data) i and j in the inferred interaction network. See section "Inferring the interaction network" below for details.

- **Ecological Network Balance Index (ENBI)**: caculated as

$$\text{ENBI} = \rho_{\text{inferred}}^{\text{D}} - \text{Median}(\rho_{\text{inferred}}^{\text{H}}) \tag{S12}$$

where ρ_{inferred} are the unhealthy (dysbiotic, $\rho_{\text{inferred}}^{\text{U}}$) and healthy (healthy, $\rho_{\text{inferred}}^{\text{H}}$) inferred net interactions for the different cases analyzed (diseases and model).

- **p-value:** We assessed statistical significance using the Mann-Whitney U test (two-sided).

- **Effect size:** We quantified the effect size of pairwise comparisons using Cliff's delta, a non-parametric measure that captures the degree of overlap between two distributions. It is defined as:

$$\delta = \frac{\text{Number of times}\left(x_i > y_j\right) - \text{Number of times}\left(x_i < y_j\right)}{mn} \tag{S13}$$

  where $x_i \in X$ and $y_j \in Y$ are observations from two groups of sizes m and n, respectively. The numerator counts the number of times a value from group X is greater than or less than a value from group Y. A value of $\delta = \pm 1$ indicates complete separation between the distributions (i.e., all $x_i > y_j$ or all $x_i < y_j$), while $\delta = 0$ corresponds to complete overlap.

  Effect sizes can be interpreted as follows: (i) *negligible* for $|\delta| < 0.15$; (ii) *small* for $0.15 \leq |\delta| < 0.33$; (iii) *medium* for $0.33 \leq |\delta| < 0.47$; and (iv) *large* for $|\delta| \geq 0.47$. See (97) for further details.

Definition of healthy and dysbiotic states

To classify states beyond a simple visual inspection of the temporal dynamics (e.g. Fig. 2B), we define the **healthy state** as the configuration of the system in which the net interaction among present bacteria is negative ($\rho < 0$). Conversely, the **dysbiotic state** is characterized by a positive net interaction among bacteria in the system ($\rho > 0$).

We examined multiple alternative definitions based on other biomarkers, including combinations of the Shannon index, richness, and the number of substances or metabolites, among others.

However, the criterion above proved to be the definition most in agreement with the classification of states obtained through visual inspection.

Inferring the interaction network

Inferring interaction networks from abundance correlations is a long-standing challenge, with well-known caveats discussed extensively in the literature (38, 80, 98, 99). To address these limitations and improve robustness, we employed three complementary approaches that move beyond purely correlational metrics: two suitable for cross-sectional datasets (i.e., data from different subjects) and another one specifically devised for longitudinal datasets (i.e., data collected over multiple time points from the same subject). All three algorithms operate on relative abundance data.

- **Method 1. *FlashWeave***: This algorithm infers interaction networks from cross-sectional data using co-abundance networks. These networks are built on the principle that bacterial species with positively correlated abundances likely exhibit positive interactions, while negatively correlated abundances suggest negative interactions. Importantly, the algorithm goes beyond simple correlation by incorporating ways to filter out indirect interactions mediated by other species, and distill direct ones. See (78) for further details.

  As output, this algorithm generates a single undirected interaction network for a given set of samples. To obtain statistical insights, we used bootstrapping (or subsampling) on each dataset. The latter involves generating multiple subsamples from the full dataset and inferring the interaction network for each subsample. For instance, Fig. S16 shows the inferred net interaction, ρ_{inferred}, for 500 bootstrapped subsamples across all datasets, including both model outputs and various disease datasets, using fixed fractions of the total sample set. Additionally, Fig. 3 displays the results for a specific subset obtained with a fraction of 0.8. Across all cases, we consistently observed that the net interaction of the inferred network is higher in diseased (dysbiotic) states compared to control (healthy) states for both real datasets and the model, respectively.

For the datasets corresponding to different diseases (IBD, IBS, CDI, CRC), we used the *sensitive* and *non-heterogeneous* options of FlashWeave in conjunction with its default parameters (78). For the data generated with the model, where the number of samples exceeds one thousand, we applied the *heterogeneous* option, recommended for large datasets in the original publication.

- **Method 2. *BEEM-Static***: This algorithm, which we used as an additional validation of the results obtained with method 1, infers ecological interactions from cross-sectional data by estimating the parameters of a generalized Lotka-Volterra (gLV) model. *BEEM-Static* explicitly models species interactions by assuming that microbiome communities are at or near equilibrium. Specifically, the method employs an expectation-maximization algorithm to iteratively estimate both species' biomasses and interaction strengths, filtering out samples that violate the equilibrium assumptions. Notably, this approach enables the inference of directed interaction networks and has been shown to outperform standard correlation-based methods in capturing both the presence and directionality of microbial interactions.

The algorithm includes various tweaking parameters, which are detailed in the original publication (79). In our implementation, we adhered to the default settings with two exceptions: we increased the maximum number of iterations within the model to 50 and applied preprocessing steps to our datasets. Specifically, we used from the datasets only bacterial taxa that were present in at least 30% of the samples and had a mean relative abundance greater than 10^{-4}, as the algorithm did not seem to work otherwise. As with Method 1, we used bootstrapping to assess statistical significance. Fig.S17 presents the results of our subsampling approach. Consistent with the findings from Method 1, the figure shows that the net interaction of the inferred network is systematically higher in diseased (dysbiotic) states compared to control (healthy) states in both real datasets and model-generated data.

- **Method 3. *LIMITS***: This algorithm leverages a discrete-time Lotka-Volterra model to infer undirected interaction networks from longitudinal data. The algorithm includes three key tweaking parameters: the number of species considered for network inference and two parameters specific to the algorithm (the error threshold and the bagging—or bootstrap aggregation—determining the number of internal iterations performed). While the inference process is relatively straightforward, we refer the reader to the original publication for a detailed explanation (80).

  For each subject we selected, among subjects with at least 20 temporal samples, a predefined number of species and infer the interaction network across various error thresholds from 1 to 5 (ranges employed in the original publication). We used the algorithm over a large number of iterations (10^6) (real data) and $5 \cdot 10^4$ (model data) to ensure robustness, although we noticed that the iteration count did not significantly affect the results. The final network was constructed by first merging the outputs of all iterations at each error threshold and then averaging across all error thresholds. In our implementation, we extended the original algorithm by computing results using both the median and the mean or average. As shown in Fig. S18, the overall patterns remain consistent, with the mean yielding slightly more stable estimates and reduced variability across replicates.

Model building

We model the gut microbiome as living in a chemostat-like environment, a common simplifying assumption (in reality, food intake and substance excretion occurs in a discrete, rather than continuous, manner). We model the ecological dynamics of the system using consumer-resource equations, also considering metabolic preferences. Specifically, we consider a number $N(t)$ of bacterial lineages (e.g., species, strains, or another suitable taxonomical group) with abundances $B_i(t)$, where i identifies the bacterial lineage. Additionally, we include a maximum number of possible resources (or substances, as we use indistinguishably the two terms here) R, with concentrations S_α, where α indicates the type of substance. Substances are introduced into the system at an inflow rate h_α and may also be generated as byproducts of bacterial metabolism. New bacterial lineages are introduced at a certain rate U with a small initial relative abundance of 10^{-4}, as long as their invasion fitness —initial per-capita growth rate of the invader in the environment created by the existing bacterial lineages— is positive. If the invasion fitness is negative and the initial biomass of the invading bacteria is sufficiently small, the new bacterial lineage is declared extinct in the next temporal step without affecting the system. Therefore, only viable invader lineages contribute to the dynamics of the system.

We explicitly incorporate metabolic pathways into the model by representing all possible routes between any two distinct substances. To describe the metabolic pathway that converts a substance S_α into another substance S_β, we introduce two key quantities:

- The energy $E_{\alpha\beta}$ = $E_\alpha - E_\beta$, or benefit, derived from the route, where E_α and E_β are the energies associated with substances α and β, respectively. This $E_{\alpha\beta}$ reflects the free energy released during the reaction, energy that we assume is fully available for growth. Nonetheless, this assumption could be relaxed by incorporating a conversion constant to account for partial energy utilization.

- The enzymatic cost, $C_{\alpha\beta}$, required to complete the pathway, which accounts for the number and complexity of the enzymes involved.

For simplicity, we explicitly focus on catabolic reactions, i.e. those in which the free energy of the reaction is positive, and account for anabolic processes indirectly only (see below). In particular, we assume that all metabolic routes where $E_{\alpha\beta} > 0$ (i.e. catabolism) are possible, and that the energy obtained and the enzymatic cost needed for the route are the same (see below for details on how this is implemented in the model via the trade-offs). Nonetheless, the framework is versatile and can accommodate more complex and realistic metabolic structures through parameter adjustments.

Each bacterial species is characterized by its own specific metabolic pathways, represented by a binary matrix P^i. In this matrix, $P^i_{\alpha\beta} = 1$ indicates that bacterium i can metabolize S_α into S_β, while $P^i_{\alpha\beta} = 0$ indicates it cannot. These pathways are randomly assigned when the bacterium is introduced into the system and remain fixed, therefore assuming no evolutionary adaptation occurs. This immigration of bacteria with random pathways constitutes the only source of stochasticity in the system.

To assign the pathways, a random number of pathways ranging between 1 and a specified maximum is first drawn for each bacterial lineage. The maximum is chosen to be relatively small to reduce computational overhead, but sufficiently large to not limit the introduction of viable bacteria into the system. We conservatively set this maximum to 10 based on the observation that, across all trade-off parameterizations considered, bacterial lineages with more than 8 pathways were unable to successfully invade and persist in the system. Then, the specific pathways are selected randomly from all possible $S_\alpha \rightarrow S_\beta$ combinations, adhering to the catabolic constraint $E_{\alpha\beta} > 0$. This approach ensures a diverse yet computationally manageable set of pathways for each bacterium, reflecting ecological variability while maintaining system feasibility.

Model equations

The equations representing the model are as follows:

$$\frac{dB_i}{dt} = B_i \left(\gamma_i \sum_{\alpha\beta}^{R} r_{i,\alpha}^{\max} \frac{S_\alpha}{K_{i,\alpha} + S_\alpha} P^i_{\alpha\beta} E_{\alpha\beta} - \delta \right) \tag{S14}$$

$$\frac{dS_\alpha}{dt} = h_\alpha + \sum_i^N \sum_\beta^R B_i \left(P^i_{\beta\alpha} r^{\text{max}}_{i,\beta} \frac{S_\beta}{K_{i,\beta} + S_\beta} - P^i_{\alpha\beta} r^{\text{max}}_{i,\alpha} \frac{S_\alpha}{K_{i,\alpha} + S_\alpha} \right) - \delta S_\alpha \tag{S15}$$

where γ_i is the yield or conversion constant from energy to biomass for bacterium i; $r^{\max}_{i,\alpha}$ and $K_{i,\alpha}$ are, respectively, the uptake rate and half-saturation constant for bacterium i feeding on substance α; δ is the outflow rate; and h_α is the inflow rate for substance α. The values for all parameters are fully described in Table S1. We assume growth parameters, r^{max} and K, to be the same across all bacterial types and substances, and that only the most energetic substance has a non-zero inflow rate ($h_0 \equiv h$; $h_\alpha = 0$ for $\alpha > 0$). For convenience, we express the (unique) energy associated with a substance α with different units, using the conversion $e_\alpha = E_\alpha \frac{M_\alpha}{V_{\text{colon}}}$; we further define such energy as a fraction of a maximum energy, i.e. $e_\alpha = e_{\text{max}}(1 - \alpha/R)$.

Finally, to ensure a balanced and realistic pathway structure within the system, a trade-off is introduced, imposing disadvantages on bacteria with numerous and costly pathways. This trade-off prevents unrealistic scenarios in which a single bacterial type that shows all pathways ultimately dominates the community. The trade-off is implemented as a growth penalty proportional to the number and cost of pathways utilized by the bacteria:

$$\gamma_i \propto \gamma \cdot g\left(\sum_{\alpha,\beta} C_{\alpha\beta}\, P^i_{\alpha\beta} \right) \tag{S16}$$

where g is a decreasing function. For the latter, we have explored the following two families of functions:

$$g(x) = \frac{b}{1 + c \cdot x^\nu} \tag{S17}$$

$$g(x) = be^{-c \cdot x} \tag{S18}$$

where b, c and ν are three positive constants that help us vary the different trade-offs inside the same family, and verify the robustness of the main results.

<u>Simulation algorithm</u>

We start our simulations with only one substance (the most energetic one, which is introduced in the system at a non-vanishing rate) and only one bacterial lineage able to metabolize this substance (and thus transform it into a random, different substance). From those initial conditions, the simulations proceed according to the following algorithm:

0. Run the ecological dynamics, given by Eqs. S14-S15 until the next invasion step.
1. At each step, eliminate species that have fallen below the extinction threshold (set to be a fraction 10^{-4} of the total biomass, that is, an abundance $10^{-4}\sum_i B_i$).
2. Introduce an invader, a bacterial lineage whose pathways are randomly selected (see above).
3. Repeat until a selected ending time.

<u>Influence of the trade-off</u>

As shown in Fig. S3, the qualitative results of our model remain robust across different trade-off functional forms and parameterizations. Nonetheless, as expected given the nature of the trade-off (see above), certain parameterizations can lead to ill-defined systems in which either no bacteria survive or only a single species dominates.

The trade-off primarily determines the total enzymatic cost allocated by each bacterial lineage and how this cost is distributed—either across multiple low-cost pathways or concentrated in fewer high-cost ones. Fig. S3F shows the distribution of total enzymatic cost per lineage across different trade-offs. We can approximate the peak of this distribution and gain insight into its width by considering the maximization of the per capita growth rate. First, we incorporate the trade-off function γ_i in the definition of the growth rate (Eqs. S14 and S16):

$$\text{Growth rate} \propto g\left(\sum_{\alpha,\beta} C_{\alpha\beta}\, P^i_{\alpha\beta}\right)\sum_{\alpha\beta}^{R} \frac{S_\alpha}{K_{i,\alpha} + S_\alpha} P^i_{\alpha\beta} e_{\alpha\beta} \tag{S19}$$

where $e_{\alpha\beta} = E_{\alpha\beta}\,\frac{M_\alpha}{V_{\text{colon}}}$ represents the energy of path $S_\alpha \to S_\beta$ in different units (see above). Maximizing this equation exactly is not possible, because it depends on all substances in the system, which are in turn determined by the pathways of the existing bacterial community. However, assuming that each substance reaches approximately similar concentrations, we can simplify the expression to:

$$\text{Growth rate} \propto g\left(C^i_{\mathrm{T}}\right) e^i_{\mathrm{T}} \tag{S20}$$

where $C^i_{\mathrm{T}} = \sum_{\alpha,\beta} C_{\alpha\beta}\, P^i_{\alpha\beta}$ represents the total enzymatic cost per bacterial type, and $e^i_{\mathrm{T}} = \sum^R_{\alpha\beta} P^i_{\alpha\beta}\, e_{\alpha\beta}$ denotes the total energy a bacterial type i extracts from its pathways (up to a proportionality constant to maintain units). Given our assumption that the energy and enzymatic cost matrices are identical (or at least proportional), the optimal total enzymatic cost per type is given by the value that maximizes:

$$\text{Growth rate} \propto g\left(C_{\mathrm{T}}^{i}\right)C_{\mathrm{T}}^{i} \tag{S21}$$

Fig. S3F shows, for different versions of the trade-off, the right-hand side of this equation alongside the distribution of total enzymatic cost per type. As shown in the figure, the peak of this right-hand side coincides with the peak of the distribution, whose width results from the fact that pathway configurations are assigned randomly and suboptimal pathways persist in the system for some time. This width, however, is also influenced by the trade-off as, when the right-hand side of Eq. S21 is flatter around its maximum, the resulting distribution is broader as suboptimal pathway configurations are similarly efficient to the optimal one, thus allowing for greater variability in enzymatic investment strategies.

In other words, if the maximum of Eq. S21 were significantly higher than the maximum energy obtainable per pathway, only bacteria with the most energetically efficient pathway would survive, leading to an ill-defined system. To obtain non-trivial communities, the trade-off function and parameters must ensure that the peak of Eq. S21 remains below the most energetic pathway. Moreover, the closer this maximum is to the mode of the distribution, the richer the microbial community will be, as a greater number of functionally distinct yet competitive enzymatic investment strategies can emerge.

An illustrative example of unrealistic functional form for the trade-off is the following (linear) function:

$$g\left(C_{\mathrm{T}}^{i}\right) = \frac{1}{C_{\mathrm{T}}^{i}} \tag{S22}$$

This function leads to a growth rate (Eq. S21 with no internal maximum, meaning that all types with different pathway configurations are effectively neutral with respect to the cost–growth trade-off. As a consequence, no configuration is favored, and the system lacks a selective pressure toward particular strategies. Fig. S24 shows three realizations for a system with this trade-off. We observe that, due to the lack of selection pressure, once the first bacterial types colonize the system and occupy the available niches, the community composition becomes effectively static. Given that real gut microbiomes exhibit temporal variability (100), this trade-off is biologically implausible.

Model simulation details

We integrated numerically the model equations (S14-S15) by using a multi-step variable-order integration scheme available in Python ($solve\_ivp$), in which we adapted the integration time steps to prevent negative abundances. The parameters used for all simulations (unless otherwise specified) can be found in Table S1. The enzymatic cost matrix, $C_{\alpha\beta}$, is assumed to be identical to the energy matrix, $E_{\alpha\beta}$, as differences are inherently accounted for through the non-linear trade-off function (see Eqs. S17 and S18). Unless otherwise specified, the simulations presented in the main text used a trade-off function defined by Eq. S17 with parameters $b = 8$, $c = 6$, and $\nu = 1.1$.

Functional coarse-graining

In the model, coarse-graining is implemented by clustering substances with similar energy levels into functional groups (see Fig. S2 for illustration). Let us consider the set of all substances S_α which, as explained in the Model Equations section, is ordered by energy level. For fine coarse-graining, substances are grouped in sets of four, meaning that S_1-S_4 are treated as a single functional entity, S_5-S_9 as another, and so on. An exception is made for S_0, which is always considered separately since it is the only substance directly introduced into the system. Once substances are grouped, we consider all possible metabolic routes between these grouped entities, resulting in $n_{\mathrm{fg}} = 43$ functional groups. For broad coarse-graining, a larger grouping size of 10 is used, leading to a more aggregated representation of metabolic pathways. In this case, the number of functional groups is reduced to $n_{\mathrm{fg}} = 9$.

In the IBD dataset, we annotated and grouped metabolic pathways according to the BioCyc ontology classification (101). We assigned a different color to each functional group, with individual pathways represented by different shades within their respective groups. In Fig. 2G, these groups are ordered from bottom to top as follows: Carbohydrate Degradation (blue), Amino Acid Biosynthesis (orange), Nucleoside and Nucleotide Degradation (light green), Carbohydrate Biosynthesis (red), All Other Biosynthetic Pathways (yellow), All Other Pathways (purple), Generation of Precursor Metabolites and Energy (dark green), Cofactor, Carrier, and Vitamin Biosynthesis (brown), Fatty Acid and Lipid Biosynthesis (pink), Nucleoside and Nucleotide Biosynthesis (light blue), Cell Structure Biosynthesis (black).

Datasets

For the analysis of diseases, we used the following datasets:

- **IBD:** Metagenomic, enzymatic and metabolomic data were obtained from the Inflammatory Bowel Disease Multi'omics Database (https://ibdmdb.org/).
- **IBS:** Metagenomic and enzymatic data were obtained from (90). Metabolomic data obtained from (102).
- **CDI:** Metagenomic data obtained from (89).
- **CRC:** Metagenomic, enzymatic, and metabolomic data were obtained from (94). In Fig. 3C CRC-1,2,3,4 correspond to MP (multiple polypoid adenomas with low-grade dysplasia), Stage-0 CRC, Stage-I,II CRC, and Stage-III,IV CRC as obtained from (94).
- **F4 and M3 gut data:** Metagenomic data corresponding to (103) was obtained from https://github.com/twbattaglia/MicrobeDS.
- **Two independent healthy cohorts:** 16S rRNA sequencing data obtained from (104).
- **Healthy data with different diets** 16S rRNA sequencing data obtained from (105).
- **Model data:** Temporal analyses and macroecological pattern assessments were based on weekly sampled data from our simulations. For analyses comparing healthy and dysbiotic states, we used 1000 samples per state (or the maximum number of samples available) for each realization, with approximately 100 independent realizations included in the analysis.

All datasets were obtained in pre-processed form, as curated by their original authors. We refer the reader to the corresponding publications for detailed descriptions of pre-processing steps. For our analyses, we only normalized count data to obtain relative abundances for each sample.

Table S1: Experimental values of the model parameters

Model parameter values

Symbol	Definition	Value	References
γ	Energy to biomass conversion	1.05×10^{13} cell/mol ATP	(106)
δ	Outflow rate	$\frac{1}{24}$ h^{-1}	(107)
h	Inflow rate of most energetic substance	2 g / (L h)	(59, 108)
K	Half-saturation constant	10^{-4} g/L	(109)
r^{max}	Maximum uptake rate	10^{-10} g/(L cell h)	(109)
U	Invasion rate	$\frac{2}{3}$ invasions h^{-1}	-
e_{max}	Maximum energy per resource consumption	3 mol ATP/mol of S	-
M	Molar mass of substances	100 g/mol	-
V_{colon}	Volume of the colon	0.41 L	(110)
R	Total number of substances	30	-

Supplementary Figures

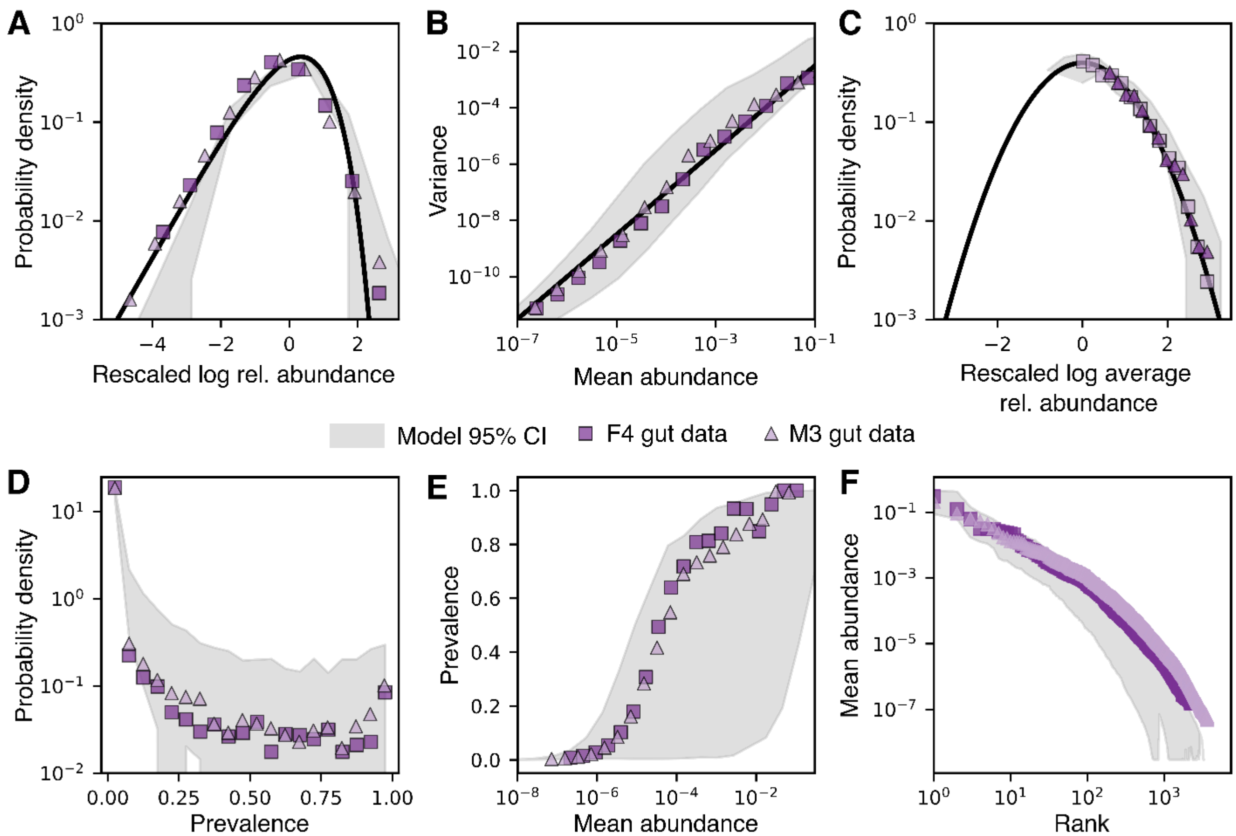

Figure S1: ***Model reproduces well-established macroecological patterns of the gut microbiome without any parameter fitting.****(**A-C**) Macroecological patterns described in (51). (**A**) The Abundance Fluctuation Distribution (AFD), which characterizes the distribution of a species' abundances across samples. The solid black line represents the expected distribution obtained by rescaling the log of relative abundance (subtracting the mean and dividing by the standard deviation) when the relative abundance follows a gamma distribution. (**B**) The mean and variance of species abundance distributions follow Taylor's Law, with variance scaling as a power of the mean, specifically with an exponent of 1.5 (black line). (**C**) The Mean Abundance Distribution (MAD), defined as the distribution of species' mean abundance across samples, follows a lognormal distribution (black line). (**D-F**) Macroecological patterns described in (52). (**D**) Distribution of species prevalence, defined as the fraction of samples in which a species is present. (**E**) Relationship between species prevalence and mean abundance. (**F**) Rank distribution of mean abundances.*

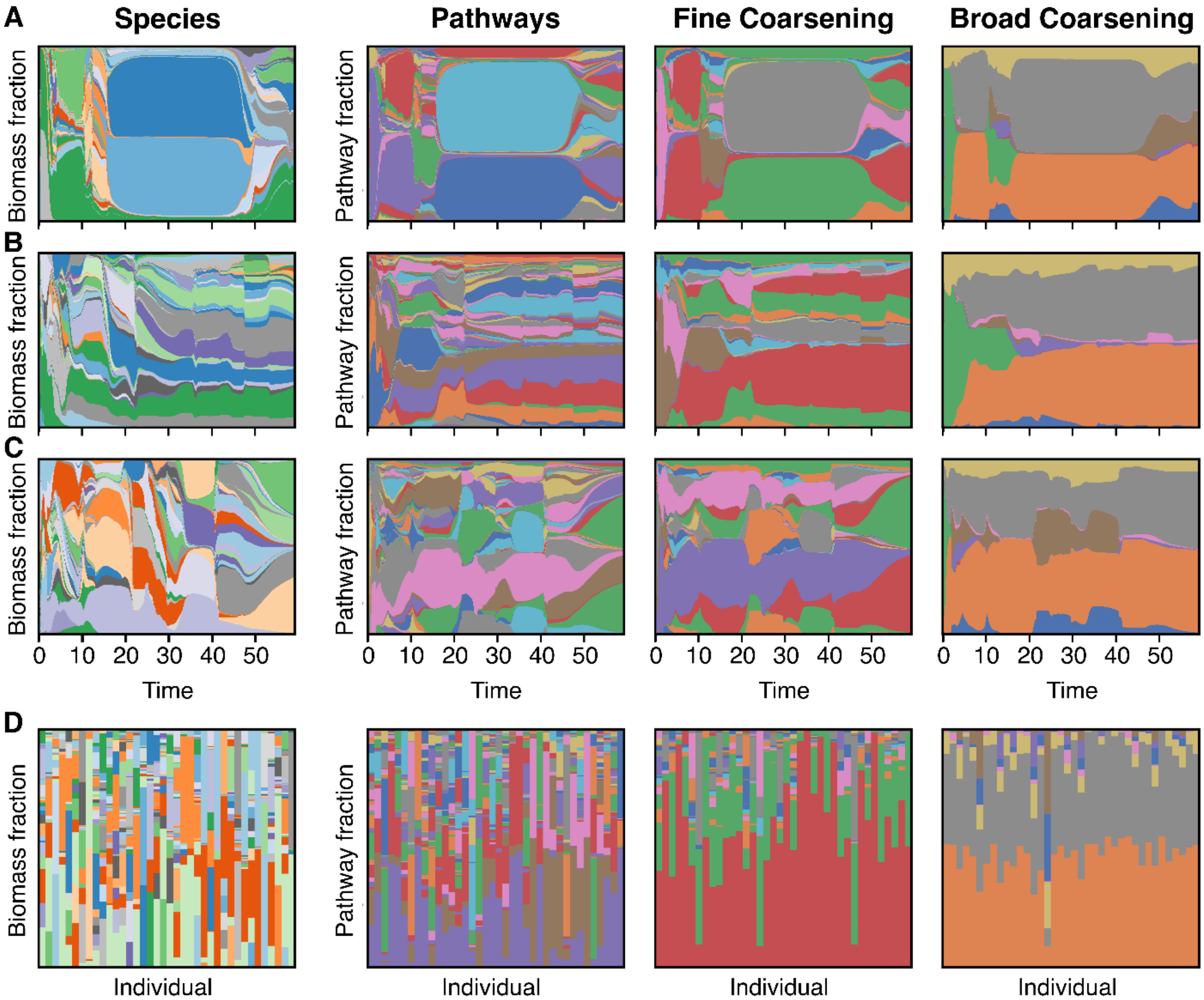

Figure S2: ***Model data reproduces functional redundancy.*** *(**A-C**) Time-series (in years) of relative species abundance (left), enzymatic cost fraction for all pathways (number of resulting functional groups, $n_{fg} = 435$; middle left), for a fine coarsening ($n_{fg} = 43$; middle right), and for a broad coarsening ($n_{fg} = 9$; right) across three different realizations. (**D**) Transversal (across individuals) evaluation of the same metrics: biomass fraction at the species level and enzymatic cost fraction under different coarsening schemes. Coarse-grained pathways exhibit functional redundancy, meaning that taxonomically distinct species share similar functional profiles (53).*

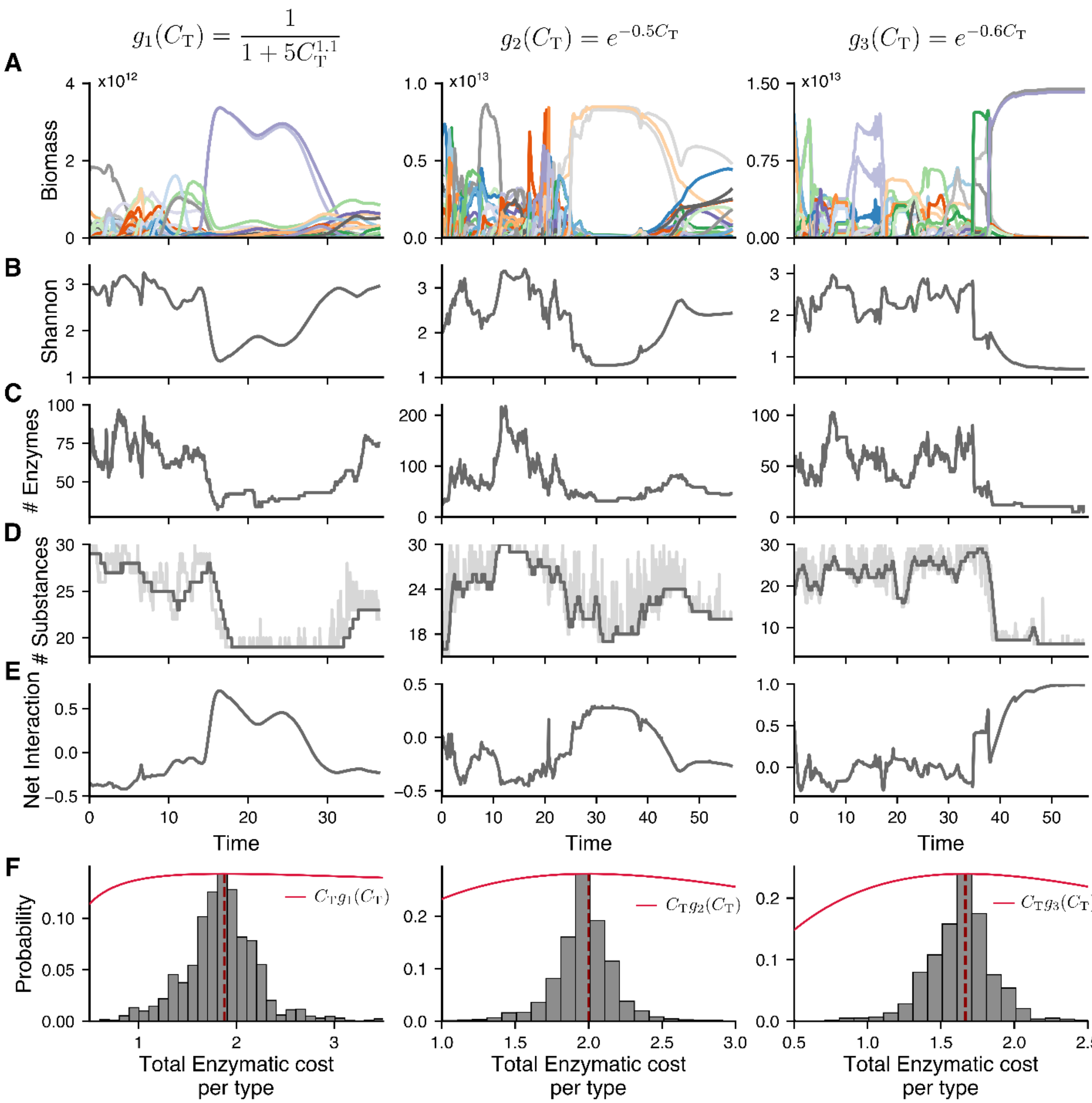

Figure S3: ***Alternative stable states are robust against trade-off changes.*** *For three different trade-off functions,* g_1 *(left),* g_2 *(middle), and* g_3 *(right), the following are shown: (**A-E**) Time-series (in years) of bacterial abundances (**A**), Shannon index (**B**), enzymatic cost (**C**), number of substances (**D**), and net interaction (**E**). (**F**) The total enzymatic cost per lineage (C_T) distribution shows a peak at the maximum of* $C_T g_k(C_T)$*, suggesting an optimal enzymatic investment strategy. The consistency of these patterns across different trade-off formulations and parameter choices indicates the robustness of the alternative stable states.*

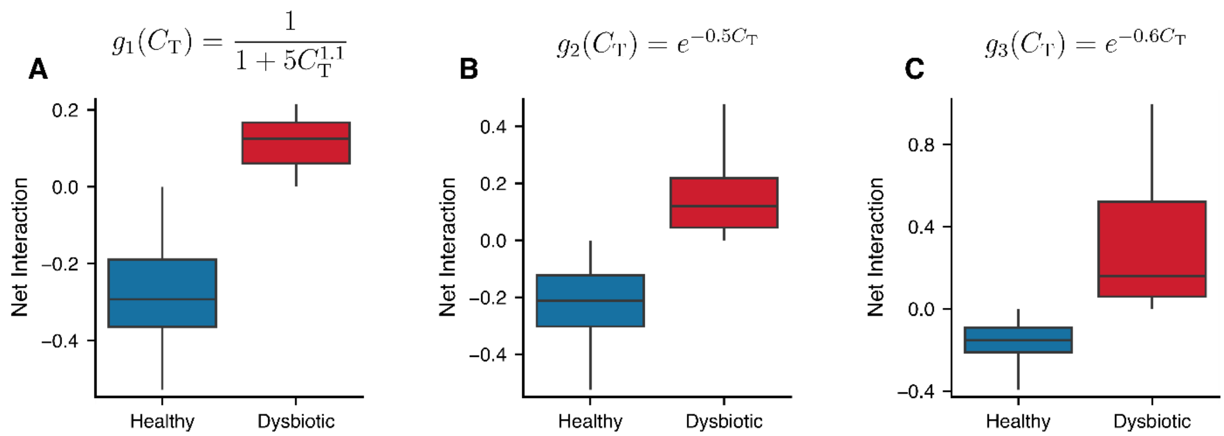

Figure S4: ***Robustness of alternative stable states to different trade-off functions.*** *Net interaction values for healthy and dysbiotic states across 10 independent simulations under three different trade-off functions: g_1 (left), g_2 (middle), and g_3 (right). Despite the different forms and parameterizations, the model consistently exhibits two distinct ecological regimes, with the dysbiotic state showing higher net interaction than the healthy one.*

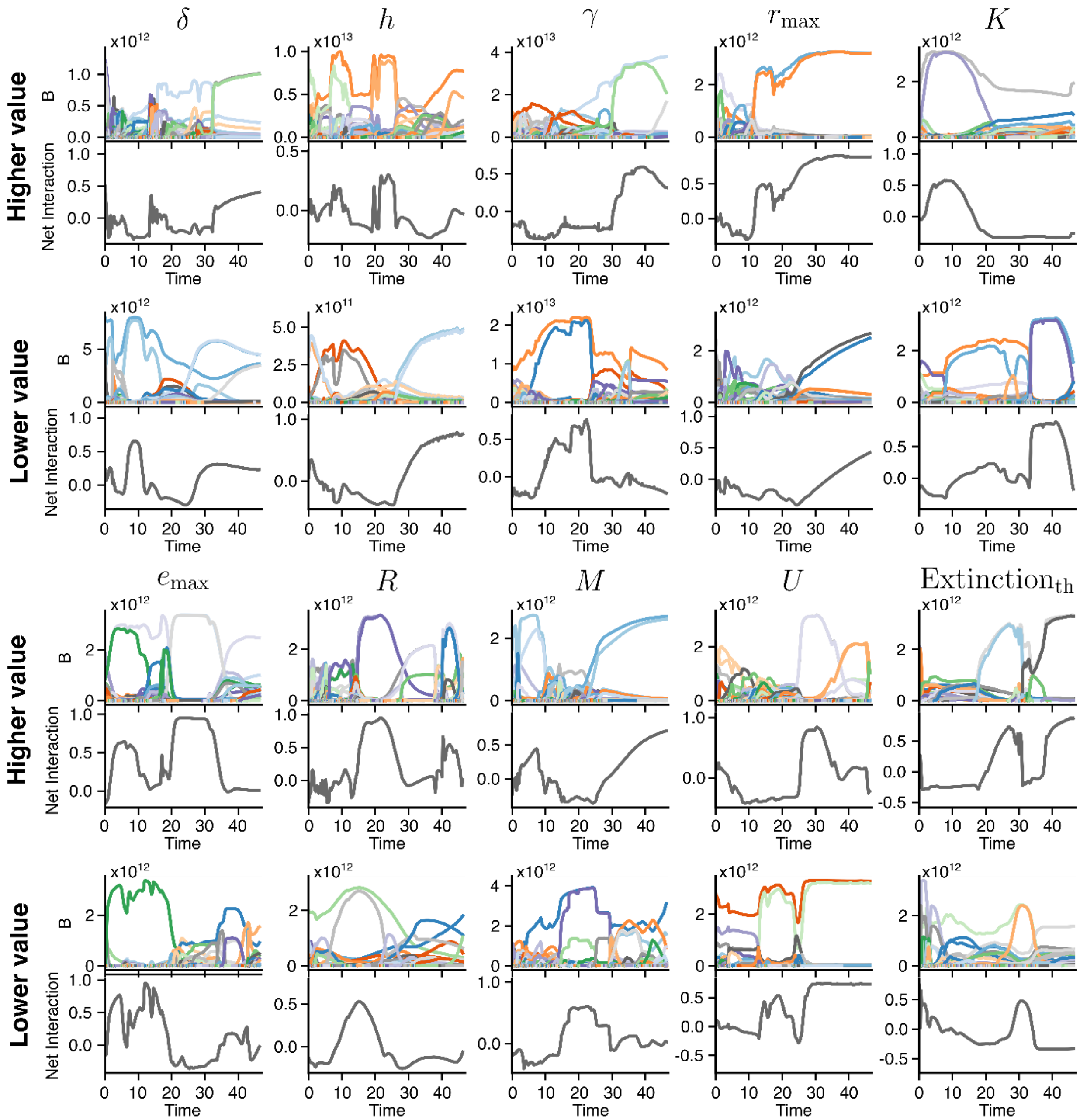

Figure S5: ***Alternative stable states are robust to parameter variations.*** *Time-series (in years) of total biomass (top row in each subpanel) and net interaction (bottom row in each subpanel) for different parameter settings in the model. Each column represents a change from the original parameter values reported in Table S1.* ***(A)*** *Increased values: from left to right, $\delta = 2.5/24$, $h = 10$, $\gamma = 1.25 \cdot 10^{14}$, $r_{max} = 10^{-9}$, $K = 10^{-3}$.* ***(B)*** *Decreased values: from left to right, $\delta = 0.4/24$, $h = 0.5$, $\gamma = 10^{13}$, $r_{max} = 10^{-11}$, $K = 10^{-5}$.* ***(C)*** *Increased values: from left to right, $e_{max} = 4$, $R = 40$, $M = 120$, $U = 4$, extinction threshold $= 5 \cdot 10^{-4}$.* ***(D)*** *Decreased values: from left to right,*

$e_{max} = 2$, $R = 20$, $M = 80$, $U = 1$, extinction threshold $= 5 \cdot 10^{-5}$. In all cases, the model continues to exhibit both healthy and dysbiotic states, demonstrating that the emergence of alternative stable states is qualitatively robust to wide parameter changes.

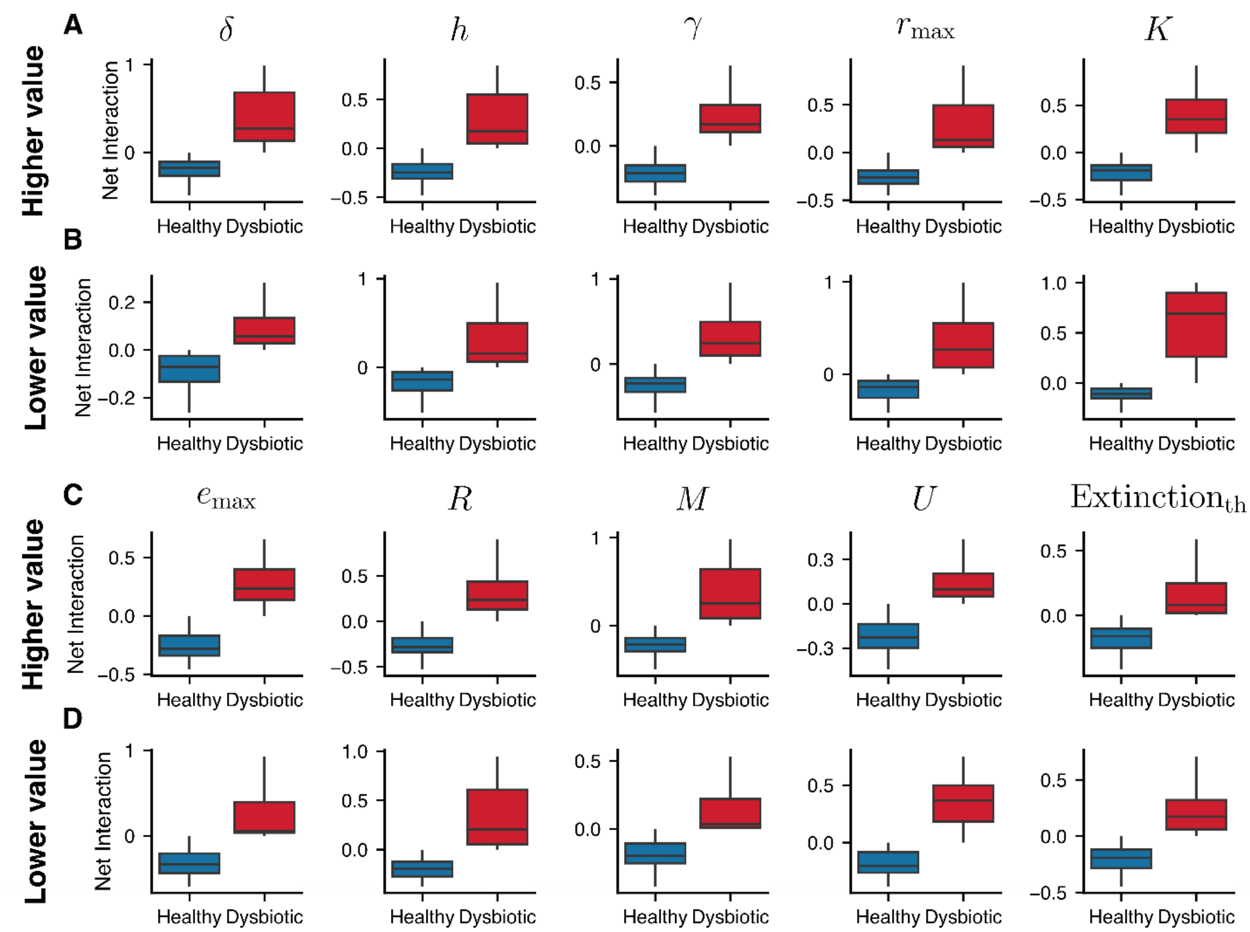

Figure S6: *Robustness of alternative stable states and dysbiosis-related increase in net interaction against parameter variations.* *Net interaction values for healthy and dysbiotic states across 10 independent simulations under variations of model parameters. Each boxplot corresponds to a distinct parameter setting, modified from the baseline values in Table S1. Higher values **(A)** $\delta = 2.5/24$; $h = 10$; $\gamma = 1.25 \cdot 10^{14}$; $r_{max} = 10^{-9}$; $K = 10^{-3}$ versus lower values **(B)** $\delta = 0.4/24$; $h = 0.5$; $\gamma = 10^{13}$; $r_{max} = 10^{-11}$; $K = 10^{-5}$. Higher values **(C)** $e_{max} = 4$; $R = 40$; $M = 120$; $U = 4$; extinction threshold $= 5 \cdot 10^{-4}$ versus lower values **(D)** $e_{max} = 2$; $R = 20$; $M = 80$; $U = 1$; extinction threshold $= 5 \cdot 10^{-5}$. In all cases, the model robustly generates two distinct ecological states, healthy and dysbiotic, with consistently higher net interaction in the dysbiotic state.*

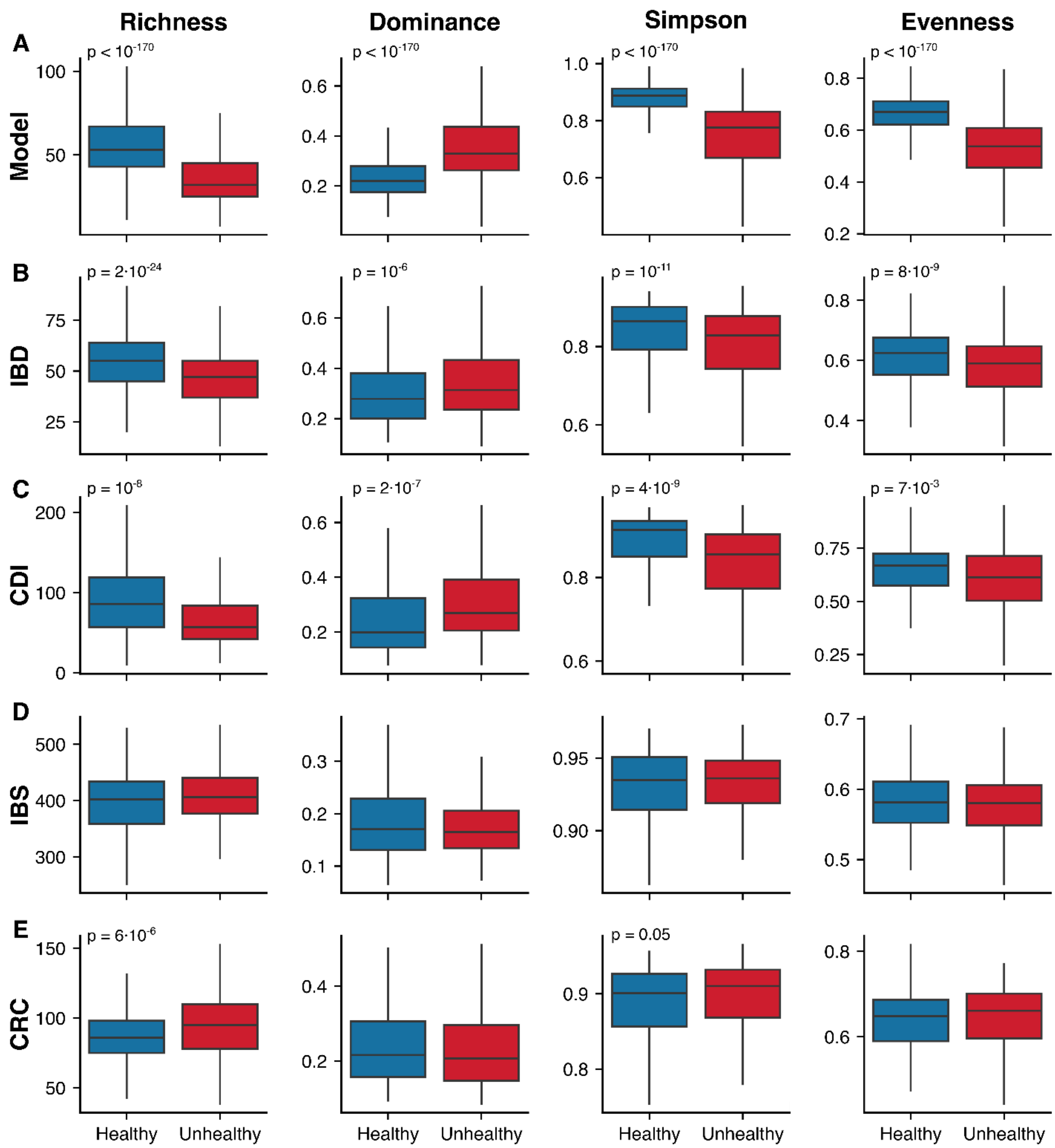

Figure S7: ***Diversity metrics in the model and real-world datasets.*** *Comparison of four diversity metrics (richness, dominance, Simpson's index, and Pielou's evenness) for the model* ***(A)*** *and four real-world datasets: IBD* ***(B)****, CDI* ***(C)****, IBS* ***(D)****, and CRC* ***(E)****. Definitions and computation details for each metric are provided in Methods (45).*

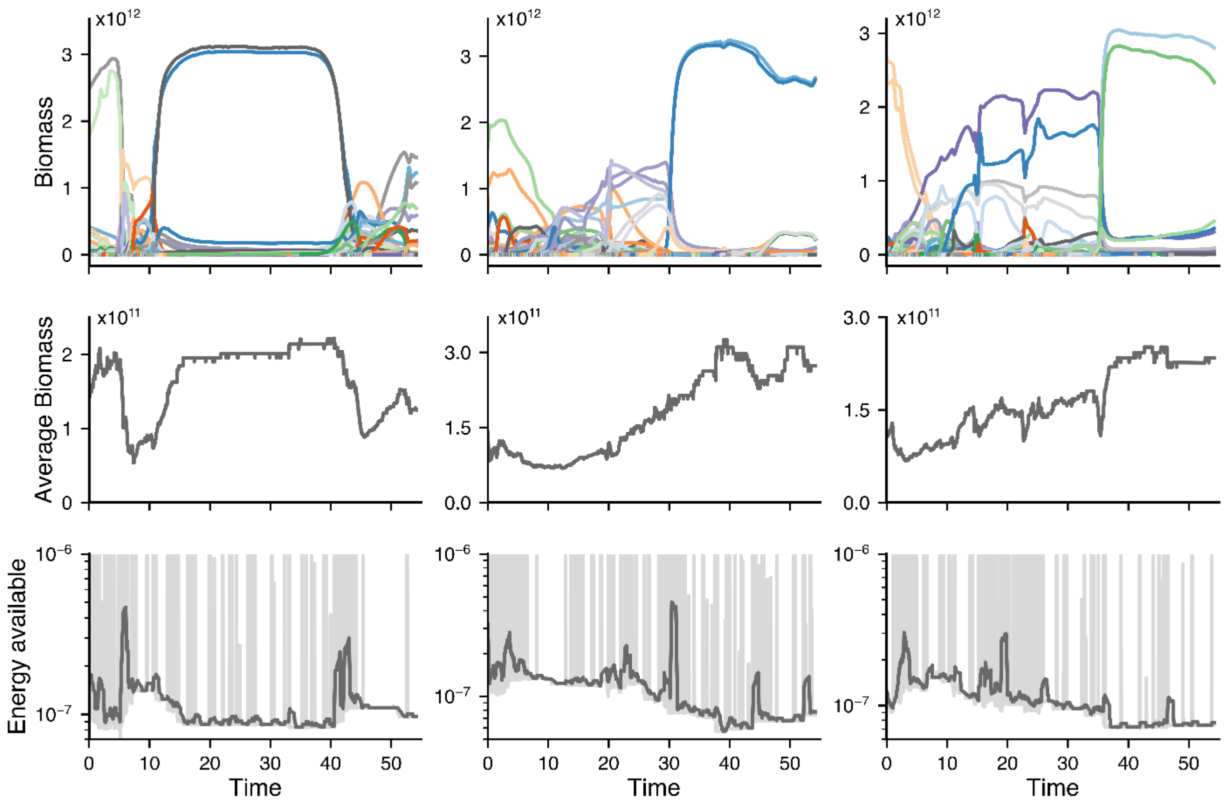

Figure S8: ***Dysbiotic state shows greater efficiency than the healthy state.*** *(**A-C**) Time-series (in years) of bacterial biomasses (**A**), average biomass across the system (**B**), and available energy within the system (**C**) for three different realizations of the model. In (**C**), dark grey lines represent the rolling average, while light grey lines depict the actual values. dysbiotic states show higher average biomass while depleting more available energy in the system, indicating a more efficient microbial community.*

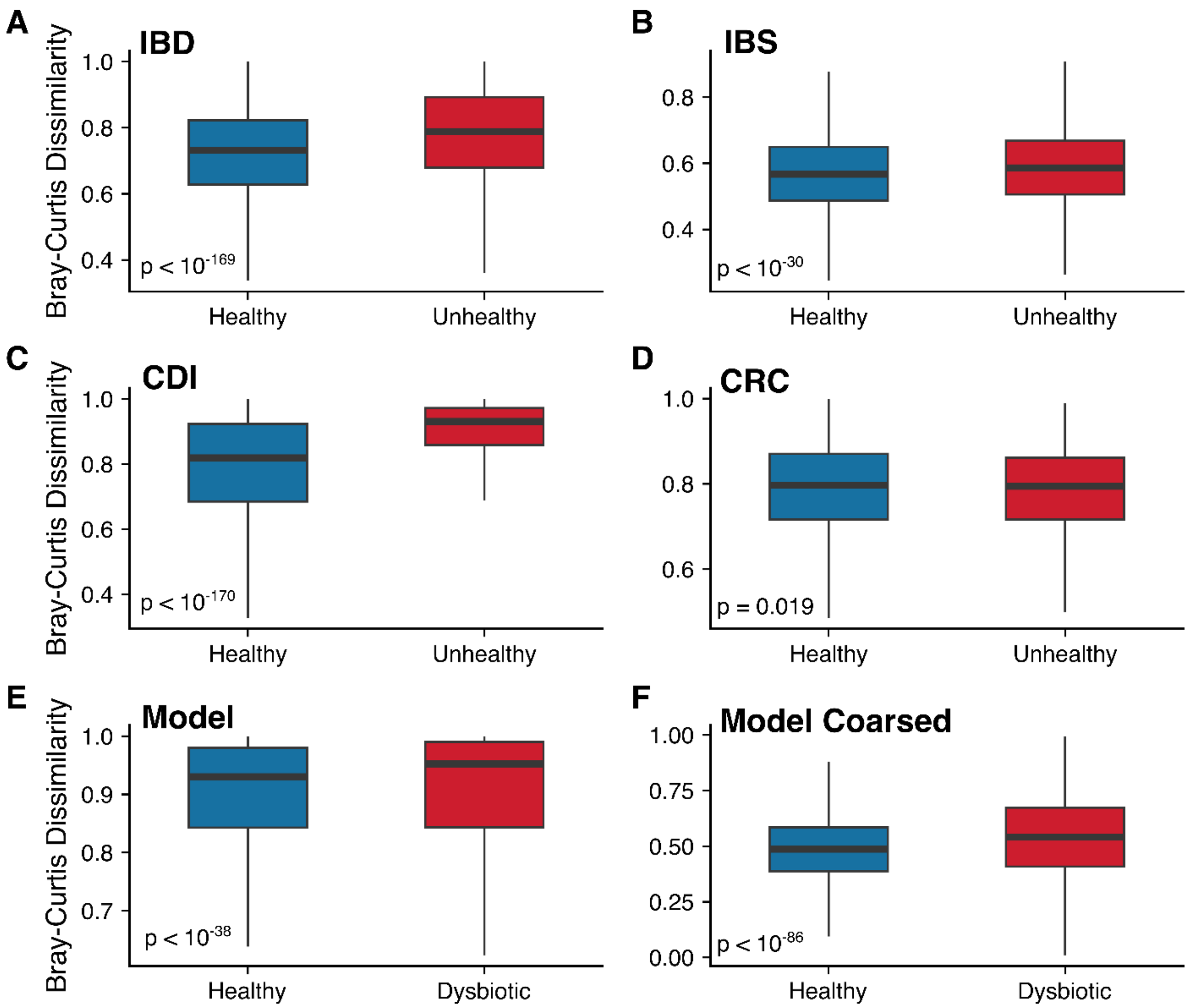

Figure S9: ***The Anna Karenina Principle holds across different diseases, but not all, and in the model.*** *Beta diversity calculated with the Bray-Curtis dissimilarity of healthy and diseased states for (**A**) IBD, (**B**) IBS, (**C**) CDI, (**D**) CRC, as well as for healthy and collapsed states in the model (**E**) and in the model with coarse-grained pathways (**F**) (see Methods (45)). Except for CRC, all diseases and the model follow the Anna Karenina Principle, which means that healthy communities are more similar to each other than dysbiotic ones. The presence of exceptions and the small magnitude of the increases in dissimilarity do not allow for strong conclusions from this observation (see (82)).* P-values (two-sided Mann–Whitney U test) were calculated as described in Methods (45).

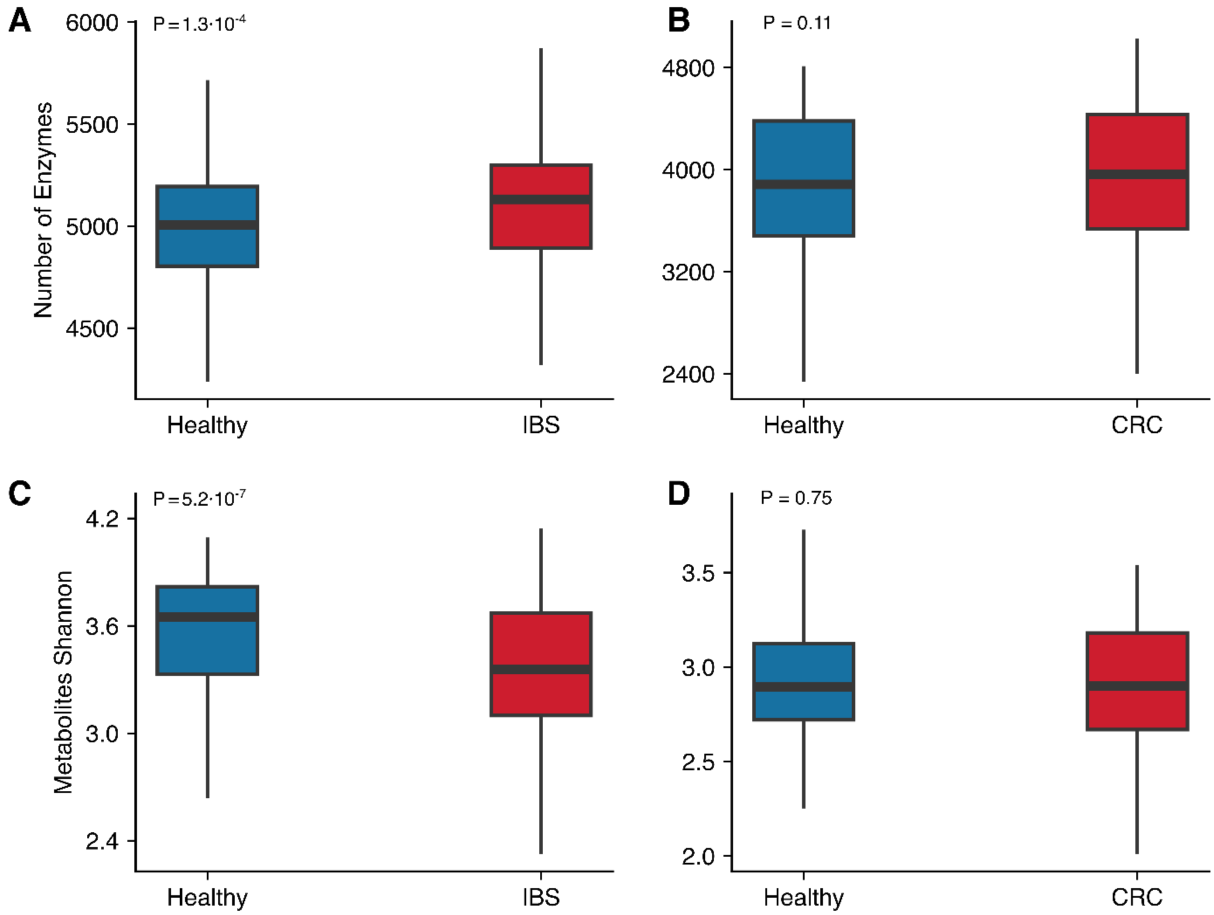

Figure S10: ***Analyzed biomarkers are not consistently reduced across diseases.*** *(**A-B**) Number of enzymes in Healthy vs. IBS (**A**) and Healthy vs. CRC (**B**). (**C-D**) Shannon index of secreted metabolites in Healthy vs. IBS (**C**) and Healthy vs. CRC (**D**). The inconsistencies observed across diseases indicate that these biomarkers alone are not reliable indicators of dysbiosis.* P-values (two-sided Mann–Whitney U test) were calculated as described in Methods (45).

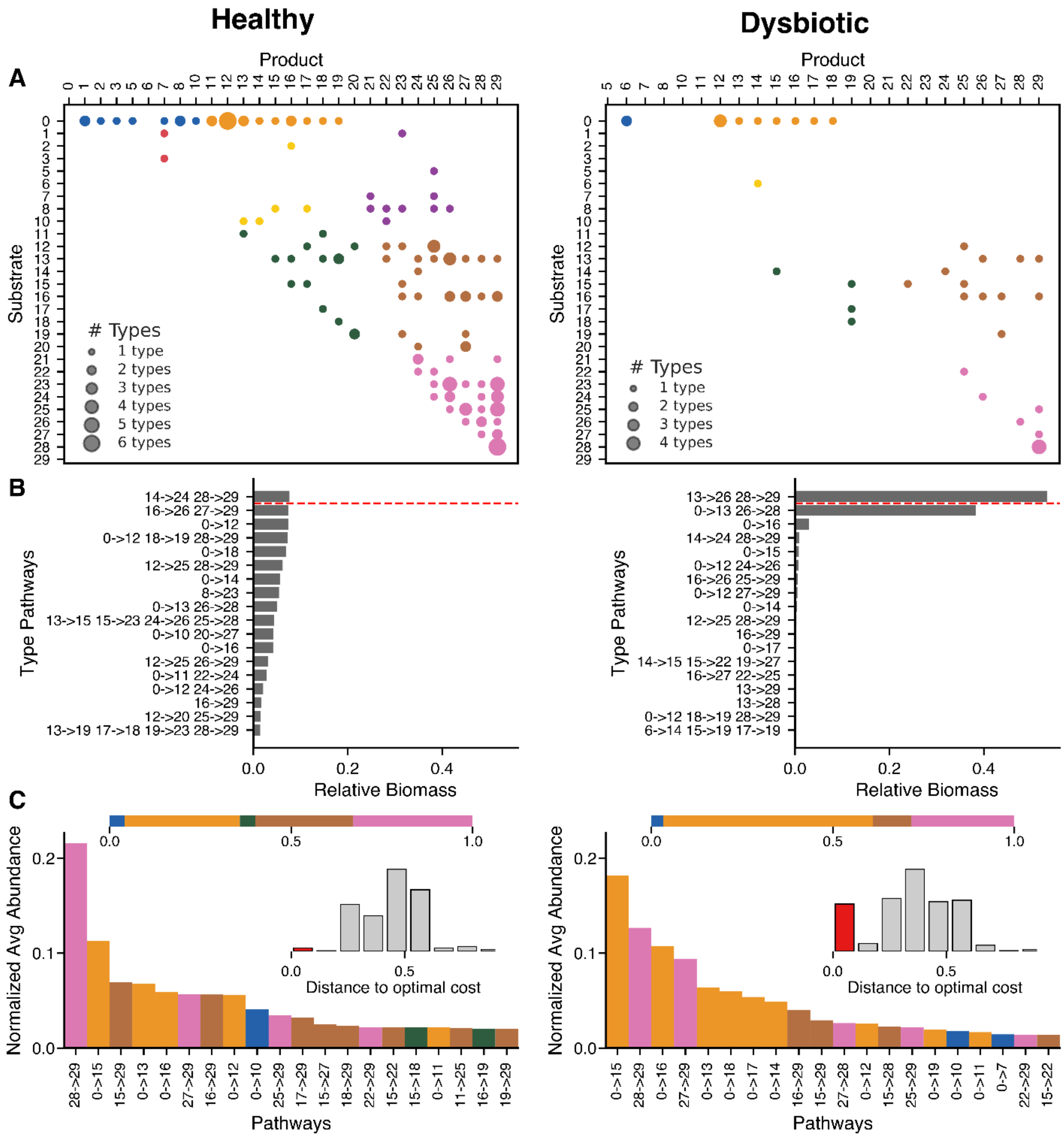

Figure S11: ***Pathway structure in healthy and dysbiotic states. (A)*** *Pathway usage matrix showing the number of bacterial lineages expressing each pathway (from substrate [row] to product [column]) at a representative snapshot of the healthy (left) and dysbiotic (right) states. Pathways are color-coded by functional category, as defined in Methods (45).* ***(B)*** *Relative biomass distribution for each pathway at the same snapshots as in panel A, for healthy (left) and dysbiotic (right) states. The red dashed line separates the 1% most abundant lineages.* ***(C)*** *Normalized average pathway abundance, focusing on the most abundant pathways among the top 1% most abundant bacterial* *lineages in each realization. Pathways are color-coded by functional category, with the horizontal bar in the inset showing the distribution of functional groups. The inset histograms depict the abundance of these top 1% bacterial lineages as a function of their distance to the theoretical optimum (given by the maximum of Eq. S18), and show that the most abundant lineages in the dysbiotic state are closer to the optimum than those in the healthy state (see red bars).*

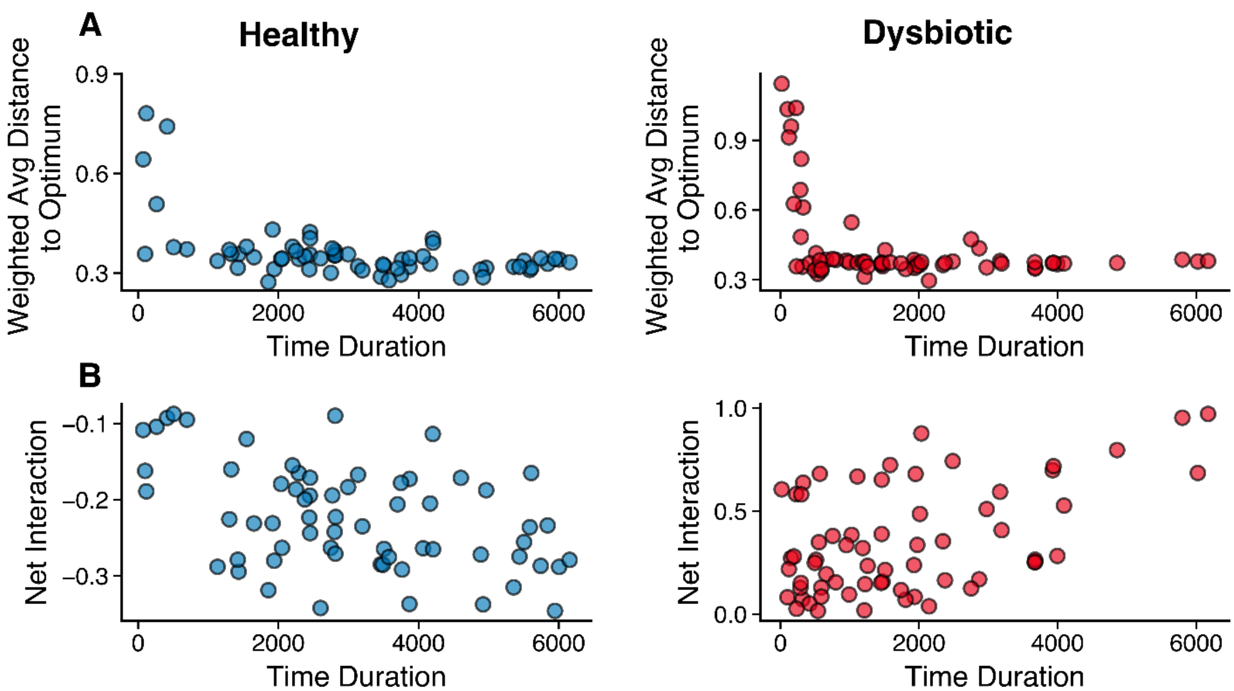

Figure S12: ***Duration of a state correlates with average metabolic efficiency and interaction structure. (A)*** *Relationship between the duration of a community state (healthy or dysbiotic) and the average distance to the optimum of the enzymatic cost–growth trade-off across its bacterial lineages, weighted by their relative abundance and duration. Each point represents the longest-lasting state for each simulation. Both healthy and dysbiotic states show a negative correlation (Spearman correlation of -0.51 with* $p - value = 1.8 \cdot 10^{-5}$ *and -0.31 with* $p - value = 0.013$*): communities composed of lineages closer to the theoretical optimum (i.e., more metabolically efficient) are more ecologically stable.* ***(B)*** *Relationship between the duration of a state and its average net interaction. In healthy states, longer durations correspond to more negative net interactions (i.e., stronger competition; Spearman correlation of -0.4 with* $p - value = 0.001$*), whereas in dysbiotic states, longer durations are associated with more* *positive net interactions (i.e., enhanced cross-feeding; Spearman correlation of 0.38 with $p-value = 0.0016$). Note that these durations (in hours), although illustrative, cannot be immediately linked to real-world timescales, since details influencing timing (e.g. initial state) differ from real-world situations.*

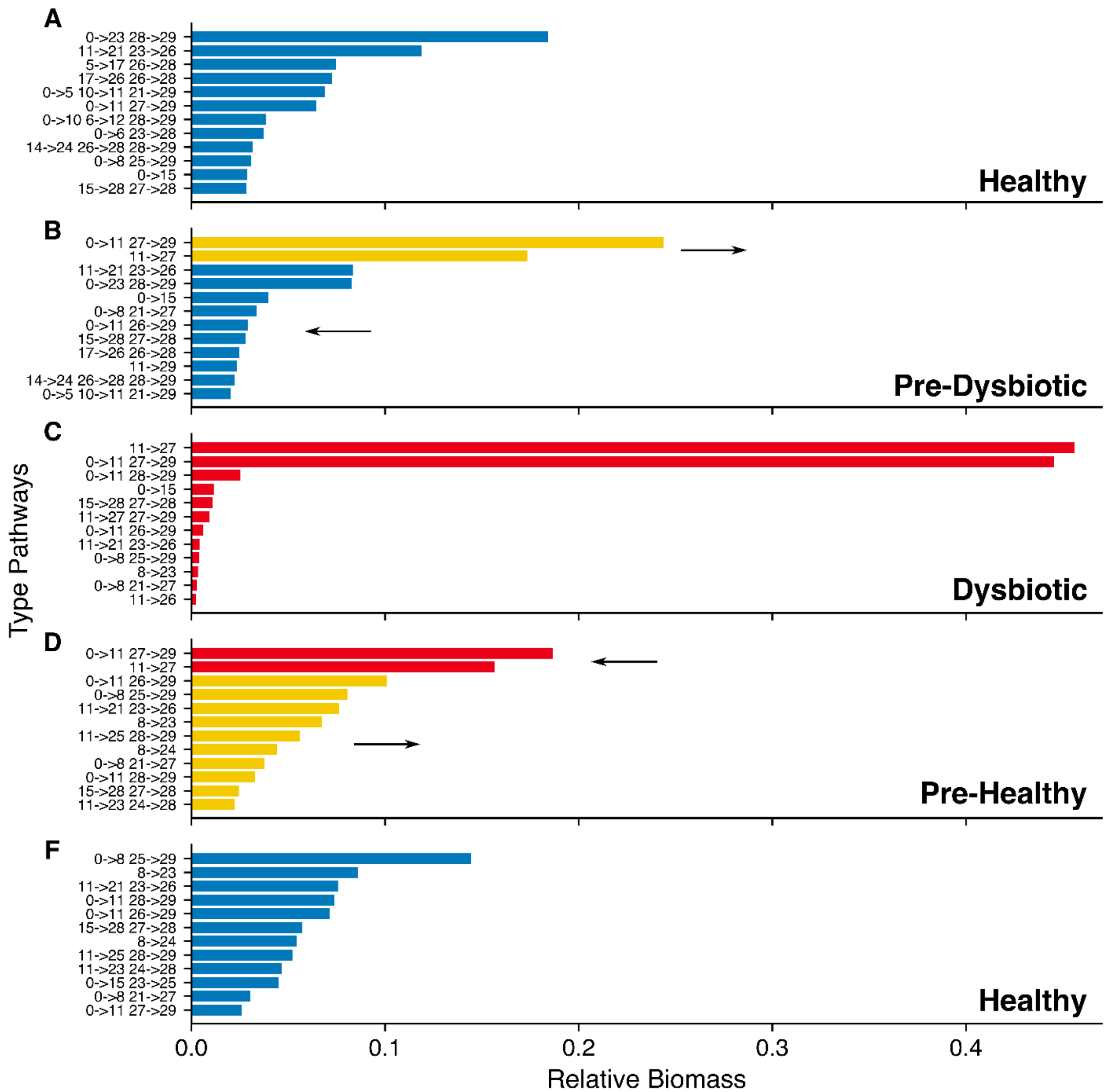

Figure S13: ***Representative example of transition between healthy and dysbiotic states.*** *Snapshots from a single realization illustrating a full transition cycle: from healthy to dysbiotic state and back. Each panel shows the relative biomass of the most abundant bacterial lineages at different stages:* ***(A)*** *healthy state,* ***(B)*** *pre-dysbiotic (i.e.,* *transition from healthy to dysbiotic),* ***(C)*** *dysbiotic state,* ***(D)*** *pre-healthy (i.e., transition from dysbiotic to healthy), and* ***(E)*** *recovered healthy state. Colors represent lineages that dominate in healthy states (blue), dysbiotic states (red), or lineages that increase during transitions (yellow). This figure illustrates how (stochastic) arrivals of specific lineages and pathways can trigger or reverse dysbiosis.*

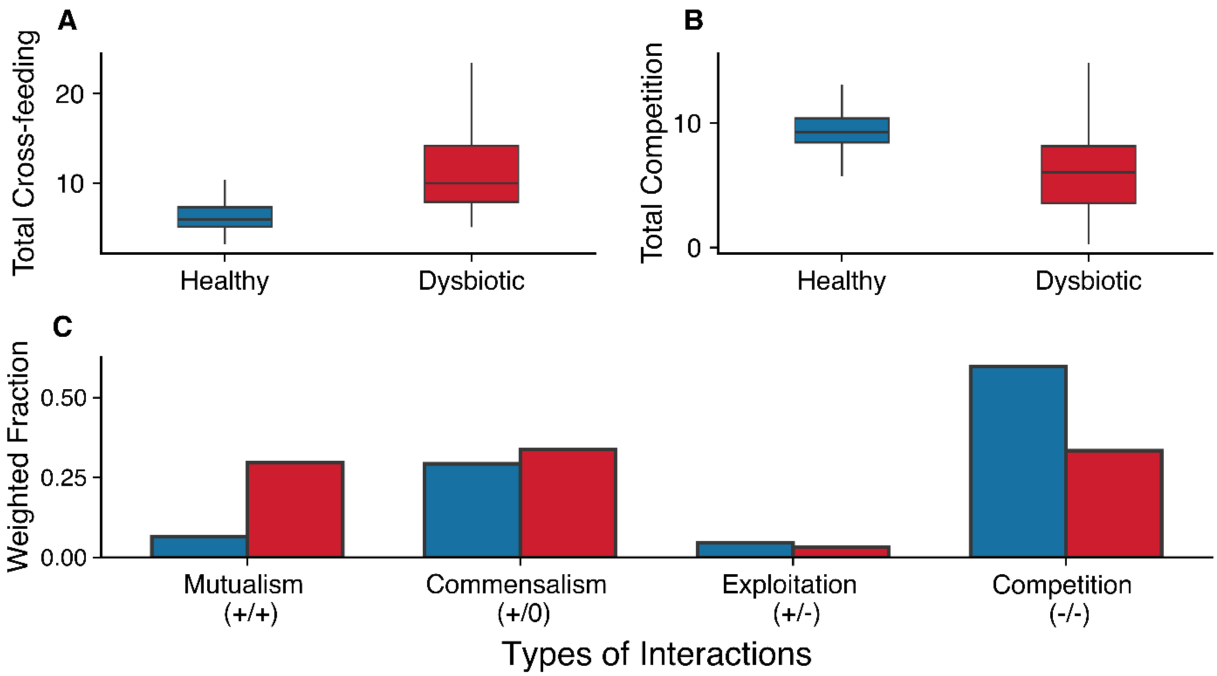

Figure S14: ***Interaction patterns in healthy and dysbiotic states in the model. (A)*** *Total cross-feeding, $c_T^+ = \sum_{ij} c_{ij}^+$, and* ***(B)*** *total competition, $c_T^- = \sum_{ij} c_{ij}^-$, where c_{ij}^+ and c_{ij}^- are computed using Eqs. 8 and S9, respectively, for healthy and dysbiotic states.* ***(C)*** *Relative frequency of interaction lineages between bacterial pairs (i, j), classified by the signs of $C_{ij} = c_{ij}^+ - c_{ij}^-$ and its reciprocal C_{ji}. Frequencies are weighted by the absolute value $|C_{ij}| + |C_{ji}|$. In agreement with real data (64) our model shows that negative interactions are predominant. Besides, diseased states in the model show an increase of positive interactions (specially mutualism) and decrease of competition.*

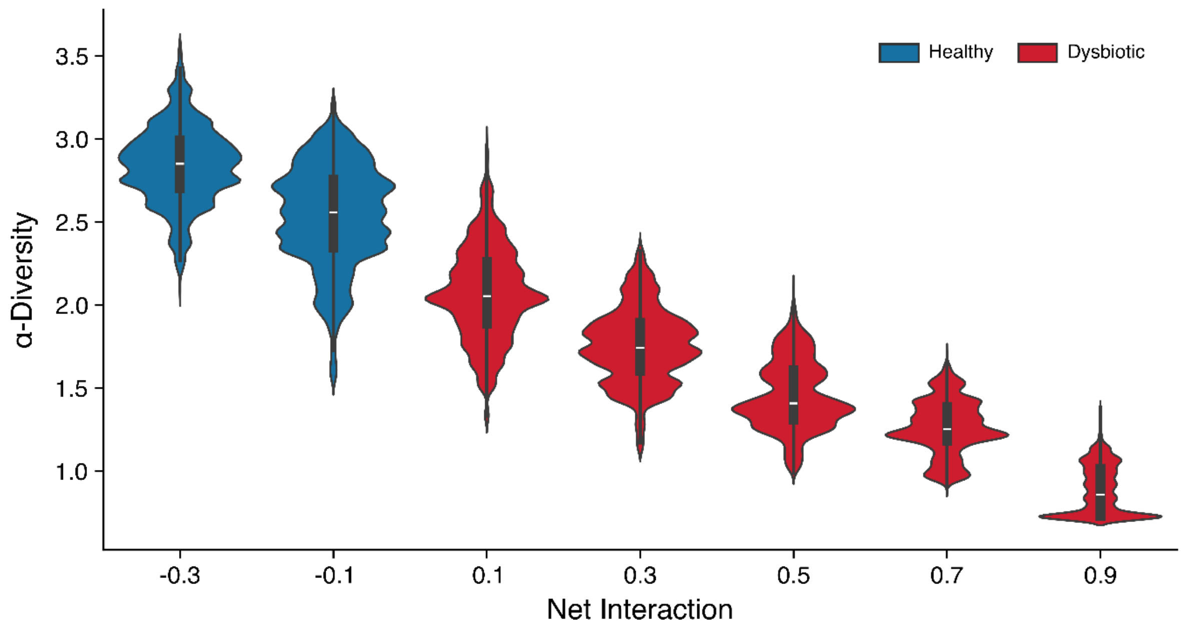

Figure S15: ***Diversity exhibits high variability across net interaction values.*** *The mean and median of α-diversity for a given value of the net interaction shows a negative correlation between the Shannon index and net interaction. Diversity is, however, widely distributed, which suggests that diversity alone is not a reliable indicator of the underlying interaction structure. For example, for the data in the figure, a value of α-diversity of 1.5 could be observed for any network with net interaction in $[-0.1, 0.7]$. Taken together, these results support the claim that a switch to a dysbiotic state (i.e. to a network with more positive interactions) typically, but not always, correlates with a decrease in diversity.*

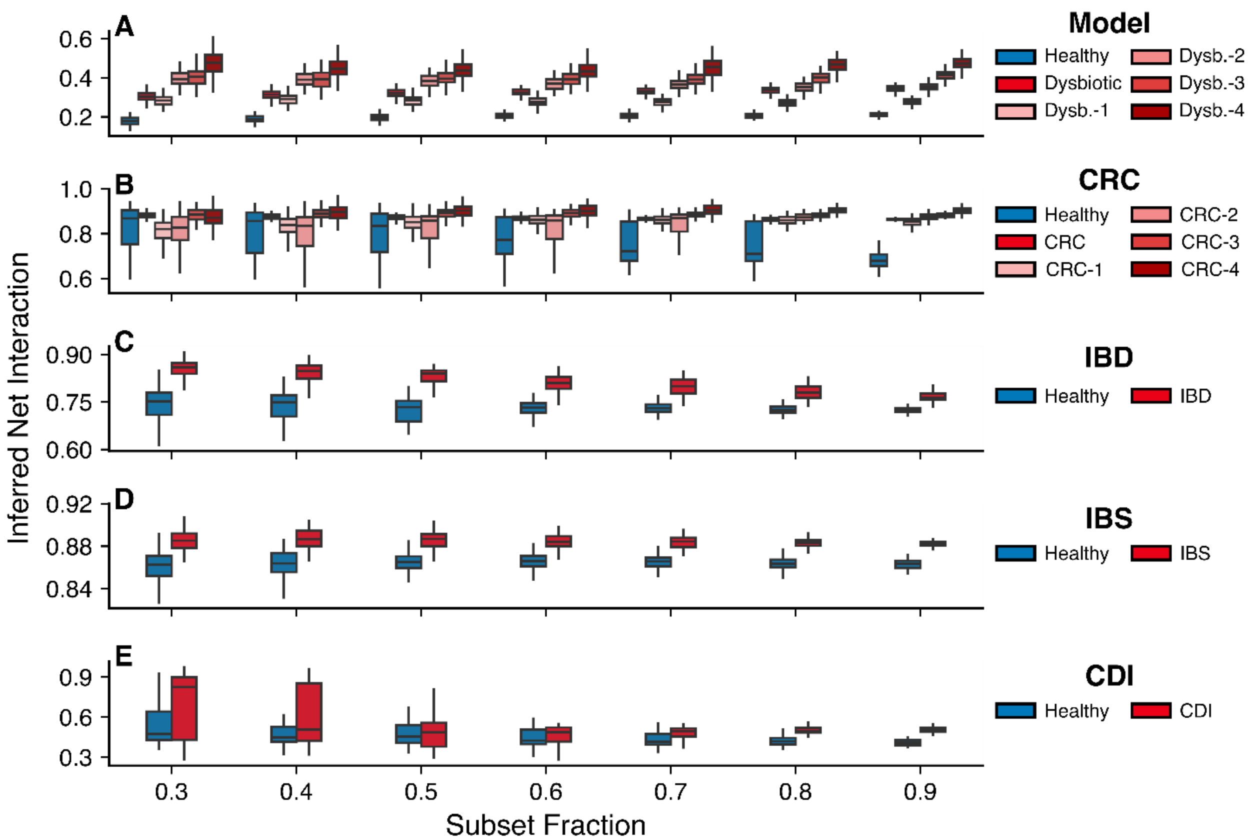

Figure S16: ***The inferred net interaction is robust under bootstrapping.*** *Inferred net interaction calculated for different subsampling fractions of the original dataset for the model (**A**), CRC (**B**), IBD (**C**), IBS (**D**), and CDI (**E**). The inferred net interaction reliably captures the qualitative trends of the true underlying net interaction, ρ, in the model across all subsampled datasets. Healthy states consistently show lower inferred net interaction compared to diseased (dysbiotic) states across all datasets. Additionally, for CRC, the inferred net interaction correlates with disease progression, with more advanced disease stages displaying systematically higher inferred net interaction across all subsampled sets. For clarity and better visualization, we opted not to include p-values and Cliff's delta values in the figure. However, all comparisons are statistically highly significant (based on 500 bootstrap-generated samples), with medium to large effect sizes as measured by Cliff's delta.*

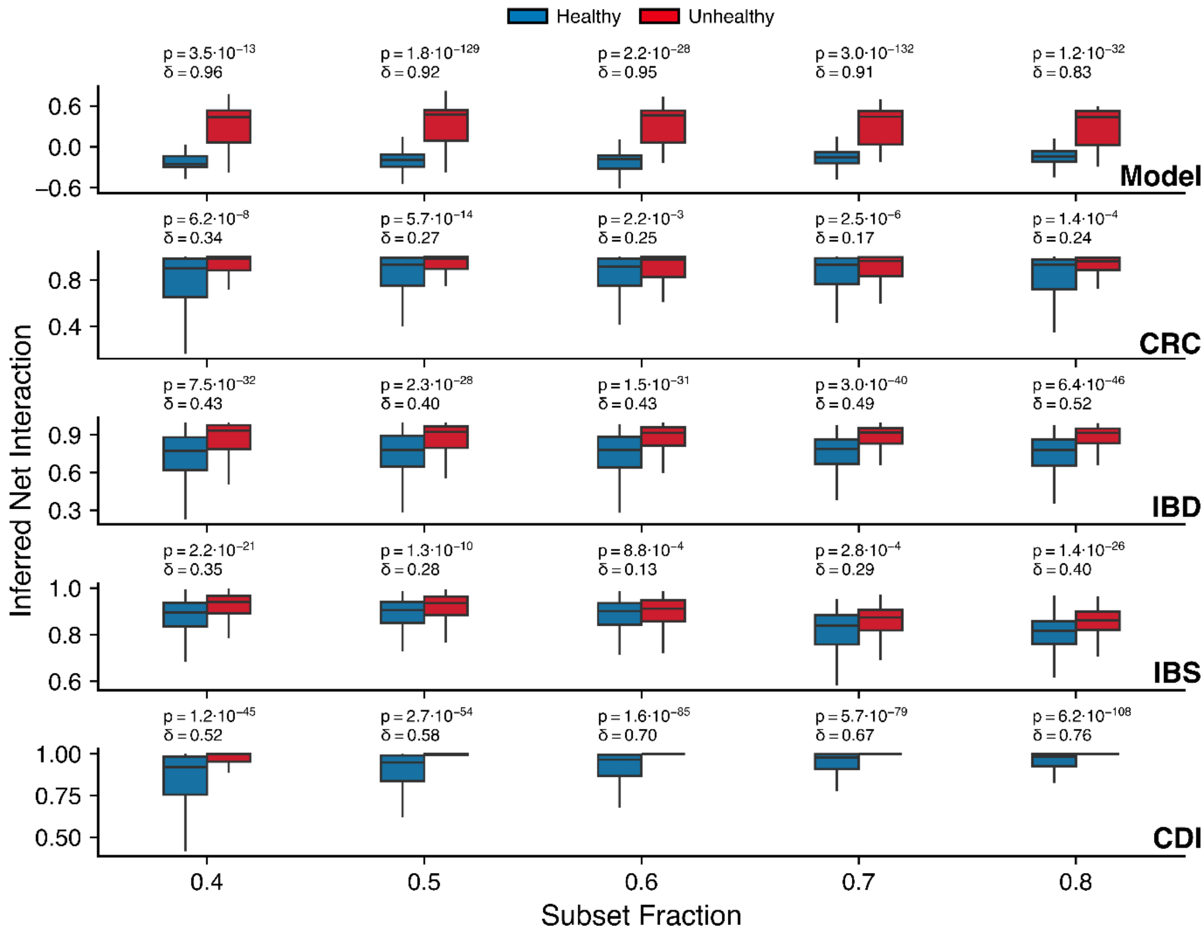

Figure S17: ***Inferred net interaction using BEEM-Static.*** *Inferred net interaction estimated with the BEEM-Static algorithm (79) (see Methods (45)) for different subsampling fractions of the original dataset for the model (**A**), CRC (**B**), IBD (**C**), IBS (**D**), and CDI (**E**). Across all cases, dysbiotic states in the model and disease states in real datasets show a consistently higher inferred net interaction, suggesting a shift toward more cooperative microbial communities compared to healthy states and healthy controls, respectively. All results are highly statistically significant (p-value calculated with two-sided Mann-Whitney U test; 500 bootstraping-generated samples) with medium to large effect sizes (Cliff's deltas).*

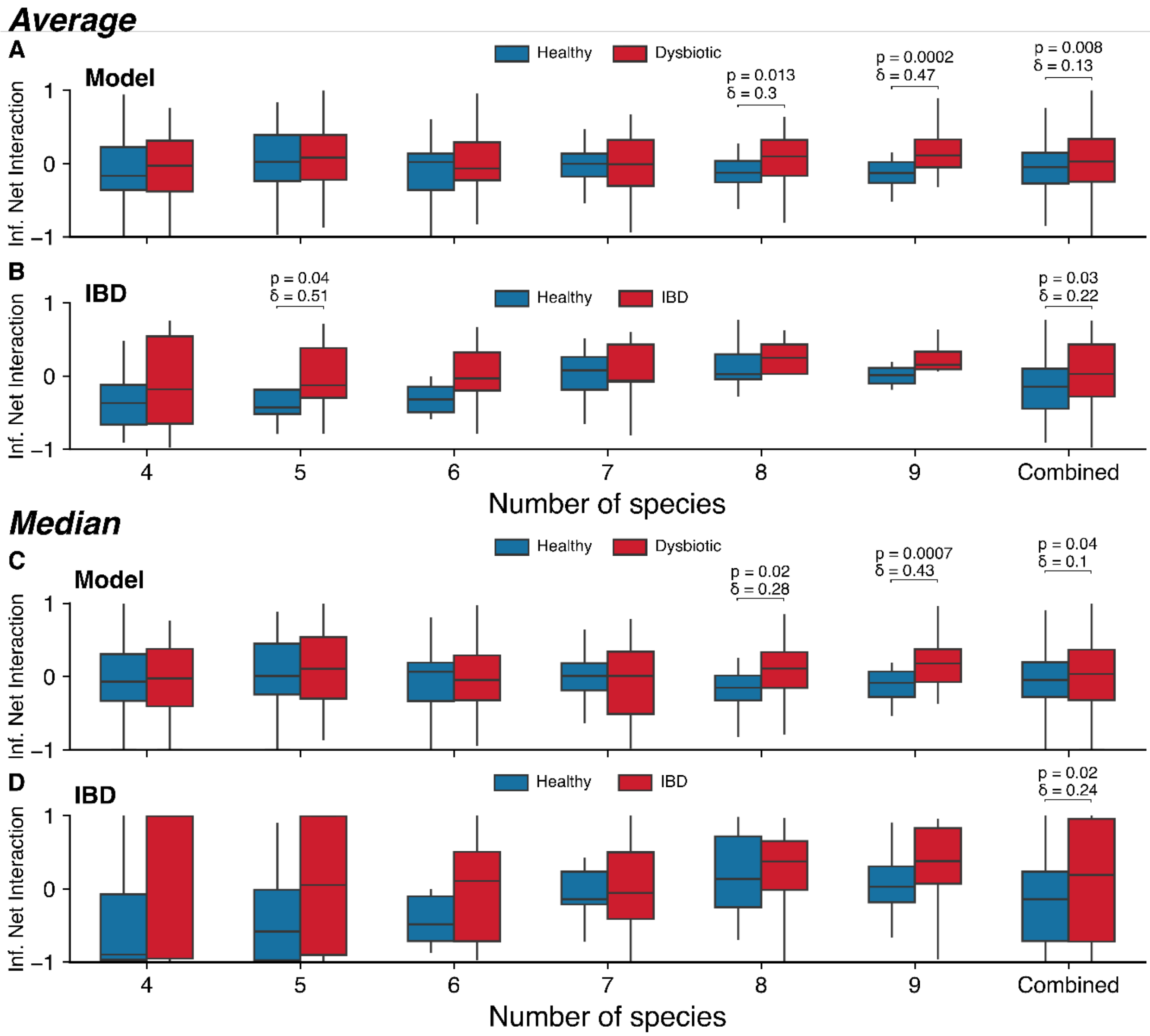

Figure S18: ***Inferred net interaction using LIMITS.*** *Net interaction inferred using the LIMITS algorithm (80) for model simulations (**A, C**) and the IBD dataset (**B, D**), using either the mean (average, top panels) or median (bottom panels) to construct the interaction networks. Either choice does not affect the conclusion: in all cases dysbiotic (model) and diseased (IBD) communities exhibit higher net interaction compared to their healthy counterparts. However, these results should be interpreted with caution, as not all comparisons reach statistical significance, likely due to the limited number of patients with at least 20 longitudinal samples.* Effect sizes (Cliff's delta) and p-values (two-sided Mann–Whitney U test) were calculated as described in Methods (45).

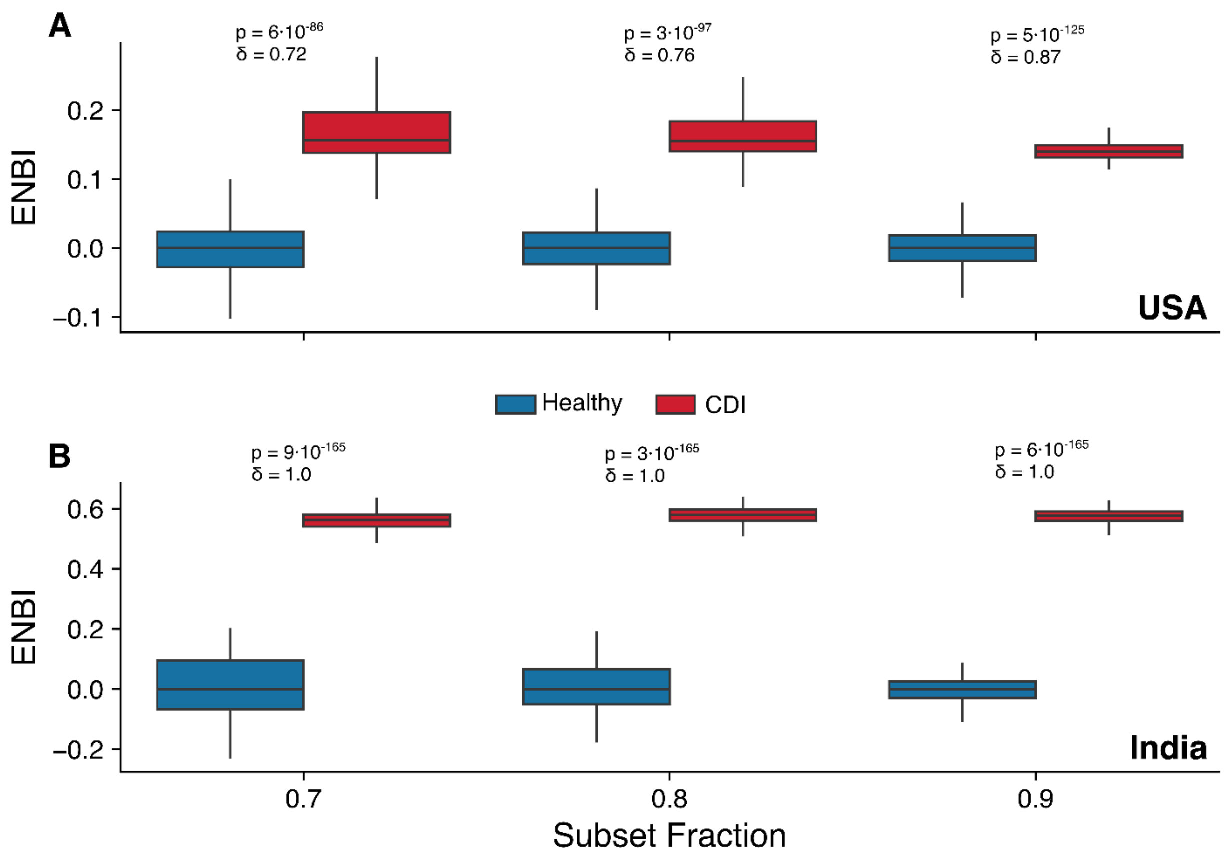

Figure S19: ***ENBI distinguishes healthy and diseased states across regions.*** *ENBI computed for Clostridioides difficile infection (CDI) patients and healthy controls from two sub-cohorts (USA* ***(A)*** *and India* ***(B)****) in the same dataset. In both regions, CDI samples show higher ENBI, indicating consistent shifts toward more positive net interactions regardless of geographic or dietary background. Effect sizes (δ, Cliff's delta) are large, indicating that the statistically significant differences (p-value calculated via two-sidedMann-Whitney U test) likely reflect biologically meaningful shifts.*

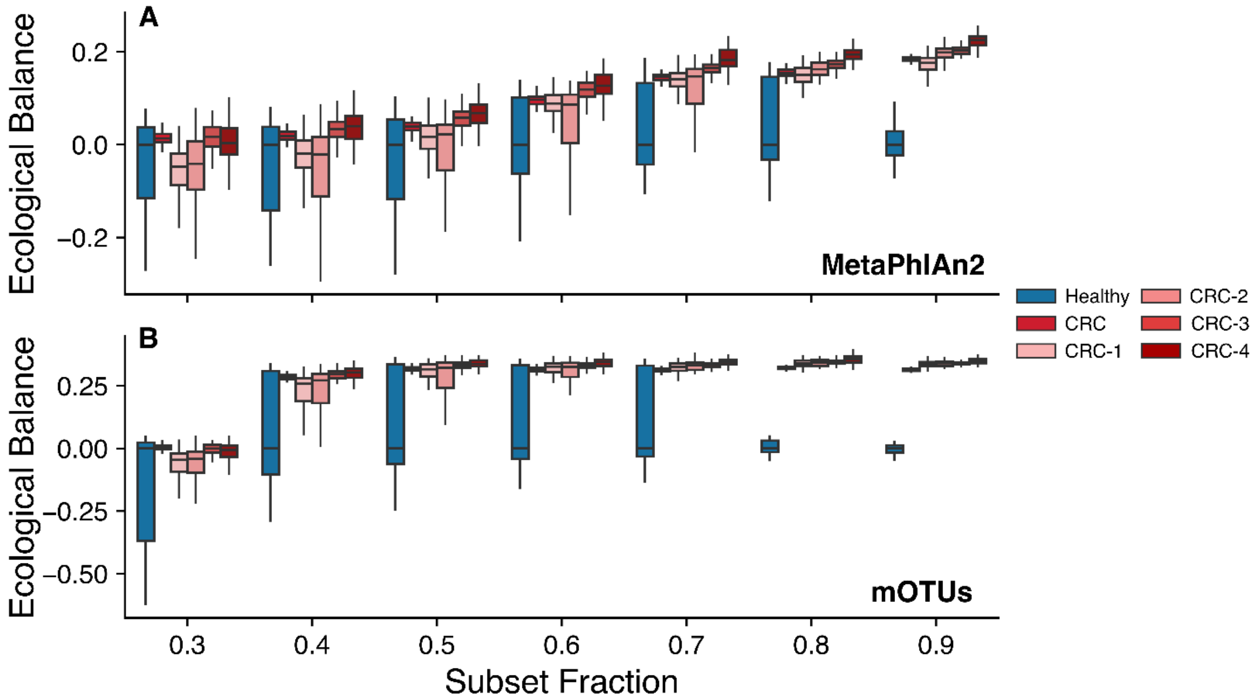

Figure S20: ***The ecological network balance index (ENBI) is robust across different taxonomic profiling methodologies.*** *ENBI calculated for different subsampling fractions of the original CRC datasets using two distinct taxonomic profiling methodologies: MetaPhlAn2 (111) (**A**) and mOTUs (112) (**B**). In both cases, disease states consistently show a positive ENBI value, indicating a shift toward more cooperative microbial communities compared to controls. Furthermore, ENBI correlates with disease progression across all subsampled datasets, reinforcing its robustness as a biomarker irrespective of the taxonomic profiling method used. All results are highly statistically significant (500 bootstraping-generated samples).*

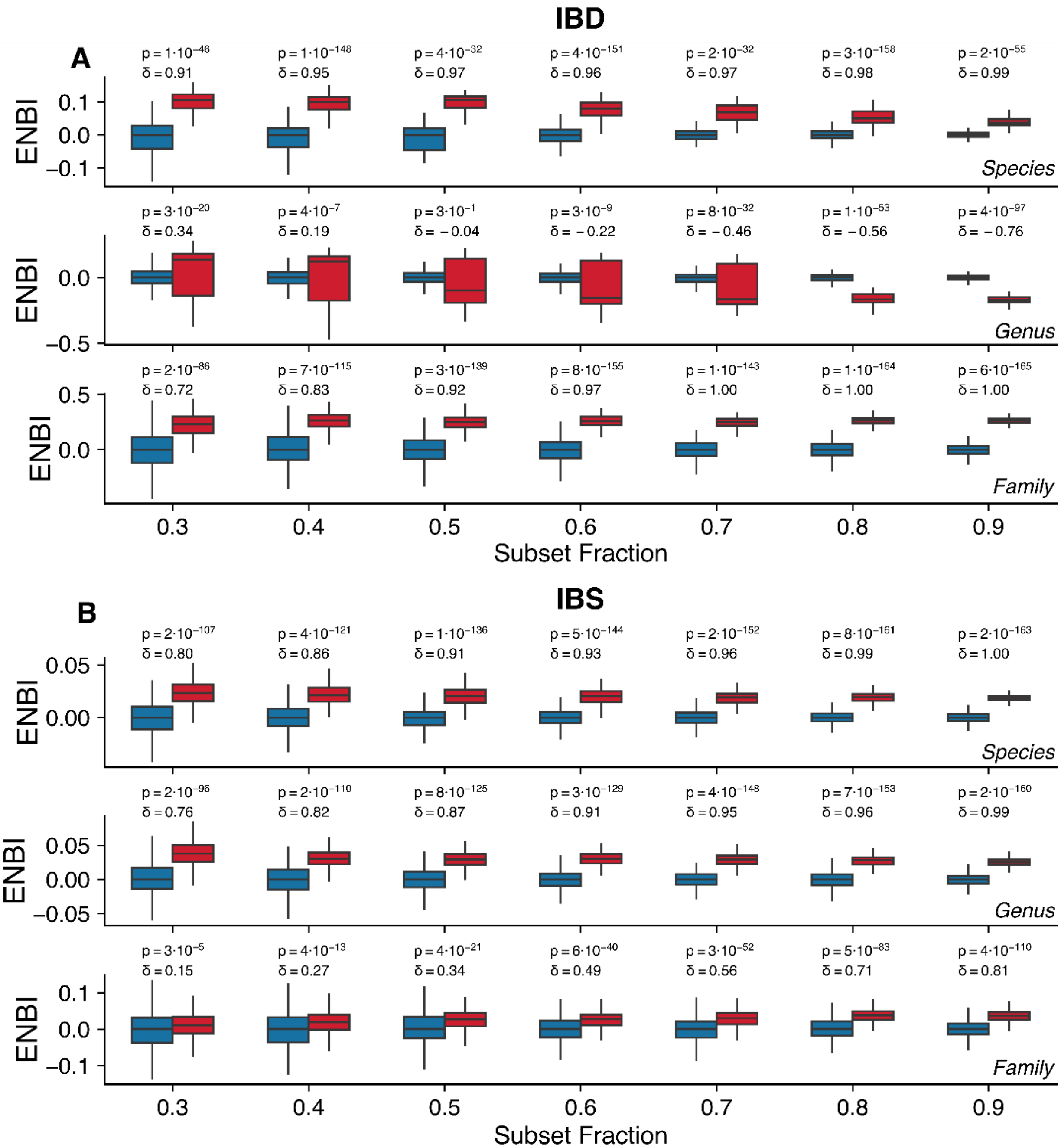

Figure S21: ***Robustness of the ENBI to taxonomic resolution.*** *ENBI computed at different taxonomic levels—species, genus, and family—for* ***(A)*** *IBD and* ***(B)*** *IBS datasets. In both cases, the ENBI robustly separates healthy and diseased states across taxonomic resolutions, with the exception of IBD at the genus level which, despite being statistically significant (p-value calculated via two-sided Mann-Whitney U test), shows smaller effect sizes (Cliff's delta), indicating a less meaningful signal.*

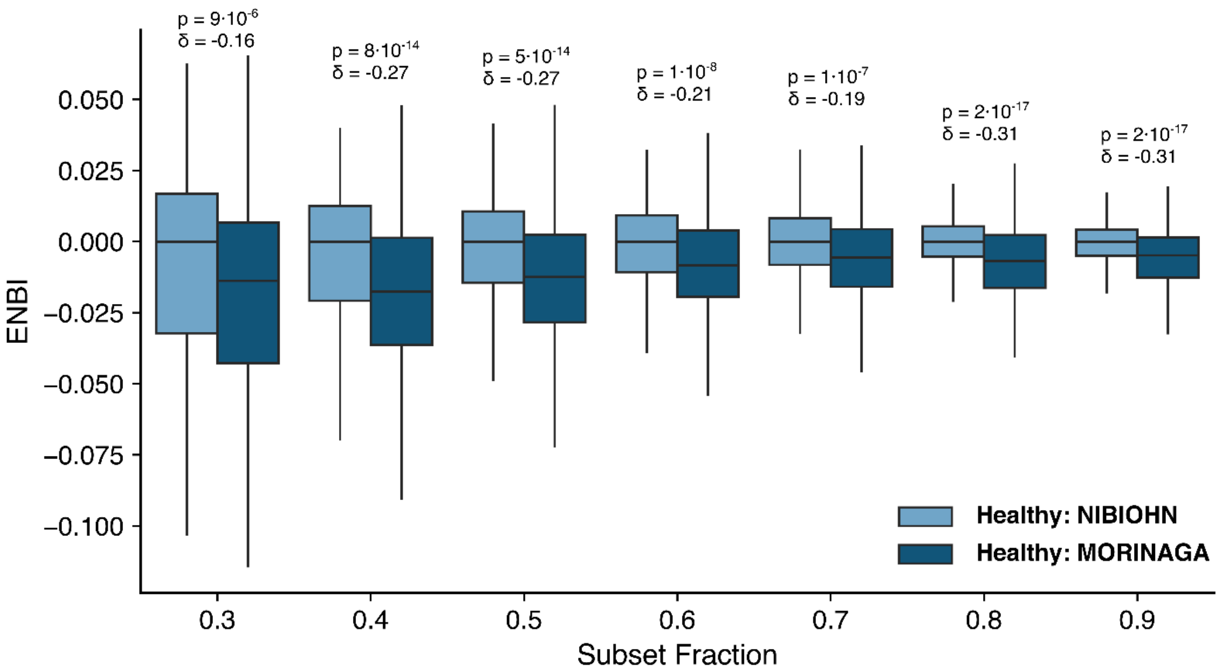

Figure S22: ***ENBI does not differentiate healthy cohorts.*** *ENBI computed for two large Japanese healthy cohorts (MORINAGA and NIBIOHN) from (104), using NIBIOHN as reference. Despite differences in geography, diet, and microbiome composition, ENBI values remain comparable, supporting its robustness. While statistical tests yield significant p-values (calculated via two-sided Mann-Whitney U test), the effect size (δ, Cliff's delta) is small, suggesting that the significance likely reflects the large sample size (500 bootstrap samples per group) rather than a biologically meaningful difference.*

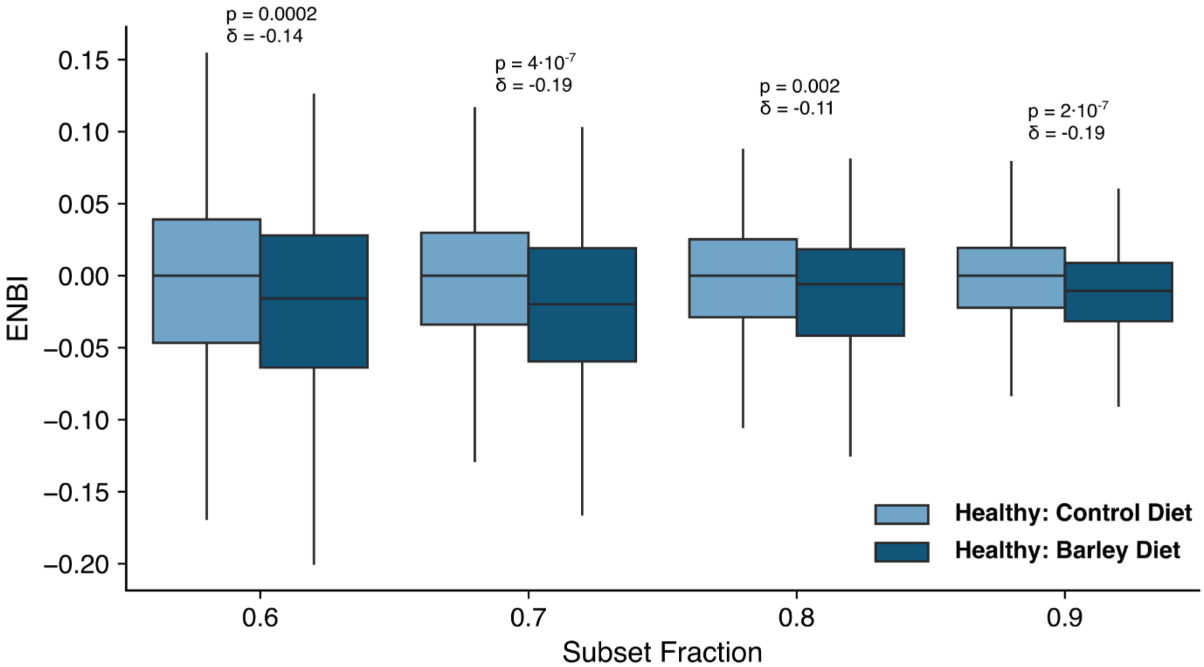

Figure S23: ***ENBI is unaffected by mild dietary interventions.*** *ENBI computed from a crossover dietary study in healthy individuals (105), comparing a barley-rich versus a controlled diet, using the latter as a reference case. ENBI values remain nearly identical, supporting its robustness to moderate diet shifts. As in the previous figure, while statistical tests yield significant p-values (calculated via two-sided Mann-Whitney U test), the effect size (δ, Cliff's delta) is small, suggesting that the significance likely reflects the large sample size (500 bootstrap samples per group) rather than a biologically meaningful difference.*

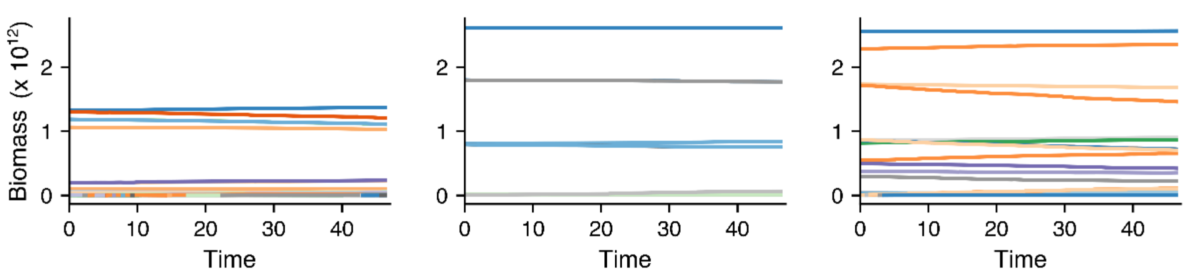

Figure S24: ***Linear trade-offs yield biologically implausible dynamics.*** *Three realizations of the model using a linear trade-off function $g(C_T) = 1/C_T$, which lacks an optimum for Eq. S21. In all cases, the system quickly becomes trapped in a single community state and exhibits negligible temporal variability, which is biologically unrealistic (100).*

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